

Program overview

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Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Master CME 2022		Master Construction Management and Engineering 2022	
CME Compulsory Courses			
AR8003TU	Legal and Governance	5	
CME1201	Collaborative Design and Engineering	5	
CME2300	Financial Engineering	4	
CME4000	Project Management	6	
CME4120	Information Systems for the Construction Industry	4	
CME4150	Managing Uncertainty and Data	4	
CME4200	Intercultural Relations	2	
CME4300	Engineering Asset Management	5	
CME Specialisations			
CME Engineering & Systems			
CME4170	Construction Technology	5	
CME4481	Systems Engineering Design	5	
CME4701	Construction Management Systems	5	
CME5030	Research and Development Methods for E&S	5	
CME Must choose at least one, but may also choose multiple			
CME4500	Engineering Systems Optimisation	4	
EPA1316A	Introduction to Data Science	5	
EPA1352	Advanced Simulation	5	
TPM024A	Methods for Risk Analysis and Management	5	
CME Projects & People			
CME2201	Dynamic Control of Projects	4	
CME4100	Process Management	5	
CME4600	Leadership and Strategic Management	4	
CME5020	Research Methodology for P&P/D&I	3	
CME Must choose at least one, but may also choose multiple			
CIE5981	Forms of Collaboration in Civil Engineering	4	
CME4800	Procurement of Complex Public Projects	5	
TPM024A	Methods for Risk Analysis and Management	5	
CME Design & Integration			
CME4100	Process Management	5	
CME4400	Entrepreneurial Engineering	5	
CME5020	Research Methodology for P&P/D&I	3	
CME Must choose one, but may also choose multiple			
AR0179	Value Capturing	5	
AR2MBE025	Urban and Infrastructure (Re)development Game	10	
TPM023A	Cost-Benefit Analysis: Theory and Application	4	
CME Ethics Courses Must choose at least 1 out of these 4 courses			
TPM003A	Water Ethics	5	
WM0312CIE	Philosophy, Technology Assessment and Ethics for CIE	4	
WM0329TU	Ethics and Engineering	6	
WM0376TU	Ethics of Technological Risks	5	
CME General Electives			
CME2100-11	Research Internship	10	
CME4900	Open Design Contractor Project	10	
CME4950	CME Research and Development Project	10	
TUD4040-10EC	Joint Interdisciplinary Project	10	
WM0201TU-ENG	Technical Writing	2	
CME Thesis			
CME5100	CME Master Thesis Preparation	5	
CME5200	CME Master Thesis	30	
CME Thesis for students who started before September 2020			
CME2000	Master Thesis	32	
CME2001	Master Thesis, Preparation	4	

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Master CME 2022

Introduction 1

For more information, please check the regulations: <http://studenten.tudelft.nl/en/students/faculty-specific/ceg/regulations-terms-and-guidelines/>

Year	2022/2023
Organization	Civil Engineering and Geosciences
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CME Compulsory Courses

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	Mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- Chao-Duivis, M.A.B., Koning, A.Z.R., Ubink, A.M., A Practical Guide to Dutch Building Contracts. 4th Edition, 2018. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%).	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning and afternoon	

CME1201	Collaborative Design and Engineering	5
Responsible Instructor	Dr. E.J. Houwing	
Instructor	Y. Liu	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In this course, students learn how to collaborate in large multidisciplinary teams on an ill-defined engineering assignment. Each group has to deal with an overload of information that needs to be structured in the design process, in order to identify the underlying engineering problem and present a solution to a client in an effective and efficient way. During the course, students are taught how to identify bad team performance and its effect on the design process, and how to intervene. Students are introduced to the field of Group Dynamics in a series of three workshops (Partnering Charter; Belbin-test; giving and receiving feedback) in the first half of the course. The outcomes of the workshops help students to organize the creation of joint deliverables: Partnering Charter, Project Management Plan, Phase I-report (analysis) and Phase II-report (design-solutions). Overall, students are exposed to an experience of intensive groupwork. Grading is based both on group and individual performance.	
Study Goals	After the course, students have gained an understanding of how to personally navigate and streamline collaborative design processes in a complex environment. Collaborative skills include: knowing how to identify shared values in a large group by making a partnering charter; how to reflect on ones personal role and behavior in a group; and how to give and receive feedback. Moreover, students learn how to acknowledge and successfully deal with different forms of disciplinary knowledge in their group. Students are thus prepared for the realities of practice where engineering assignments tend to ill-defined in the early stages of a project, and multidisciplinary collaboration is a indispensable.	
Education Method	In this course, students learn how to collaborate in large multidisciplinary teams on an ill-defined engineering assignment. Each group has to deal with an overload of information that needs to be structured in the design process, in order to identify the underlying engineering problem and present a solution to a client in an effective and efficient way. During the course, students are taught how to identify bad team performance and its effect on the design process, and how to intervene. Students are introduced to the field of Group Dynamics in a series of three workshops (Partnering Charter; Belbin-test; giving and receiving feedback) in the first half of the course. The outcomes of the workshops help students to organize the creation of joint deliverables: Partnering Charter, Project Management Plan, Phase I-report (analysis) and Phase II-report (design-solutions). Overall, students are exposed to an experience of intensive groupwork. Grading is based both on group and individual performance.	
Assessment	The final grade is based on both group and individual assignments (50/50). Assessment is based both on the quality of the joint deliverables and their presentation, online peer-assessment and the self-reflection report.	
Expected prior Knowledge	Positive result for Intercultural Relations. The course is therefore focused on disciplinary knowledge and personality as drivers for group dynamics, rather than cultural diversity.	
Academic Skills	Analytical thinking/ Writing reports/ Cooperation/ Professionalism/ Debating and discussion / Reflection on own personal and group behavior/ T-shaped engineering profile.	
Literature & Study Materials	Reader & Smulders et al. (2012) Teamworks by students, for students.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Permitted Materials during Exam	No exam is included in this course	
Collegerama	No	

CME2300	Financial Engineering	4
Responsible Instructor	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Compulsory course for master CME - Construction Management & Engineering (4TU)	
Summary	Reader Financial Engineering (pdf on Brightspace)	
Course Contents	Short overview of the theory of the firm Sources of finance, cost of finance, capital markets Cash flow models, decision rules like payback period, C/B ratio, NPV Financial accounting Financing (international) projects, project finance Political and social factors Portfolio management Financial risk management Influence of future trends on sustainability and circularity on finance This course deals with the finance issues related to the implementation of civil engineering projects. It introduces economic engineering concepts and finance-related topics such as project financing and financial accounting. This course requires the student to study in detail: # Finance and the firm, covering topics such as sources of finance, cost of finance, financial structures, working capital management and financial accounting # Capital Budgeting Decisions and Risk, covering topics such as Capital Budgeting, Political and Social factors, Portfolio Management and Risk Considerations # Project finance, covering topics such as international capital markets, stakeholder/actors viewpoints and cash flow modelling of projects including characteristics of typical projects like oil wells, open mining, infrastructure and office buildings # Future trend topics for finance, especially implications of sustainability and circularity	
Study Goals	The final attainments of this course that students are able to: # Understand the financing and financial implications of civil engineering projects from both a firm and a project perspective, including perspectives from financial involved actors # Create a cash flow overview from cost engineering information # Apply investment and financing analytical techniques # Assess a project proposal and investment strategies # Apply risk analytical techniques on financial data of a project or proposal # Understand the project life-cycle and its impact on and relationship with project finance # Apply and understand the valuation of (financial) assets and projects under uncertainty. # The ability of students to deal with uncertain political and social factors and financial risks # Understand future financing and business trends with regards to sustainability and circularity	
Education Method	This course will include lectures, case studies and an individual assignment.	
Books	Principles of Corporate Finance, Brealy, Myers and Allen, Eleventh version or later, Global Edition	
Assessment	The final assessment is a written examination (40 multiple choice questions). The use of an calculator is allowed. No cell phones or tablets are allowed. Continued pandemic risks may imply an alternative examination for 2022/23. - The result of the multiple choice exam should be 6.0 or higher.	
Contact	CME Secretary's Office, CiTG room 3.40, mrs. Sandra Schuchmann - Hagman, s.c.m.schuchmann@tudelft.nl tel: +31 (0)15 27 84774	
Expected prior Knowledge	No prior knowledge is required.	
Academic Skills	To prepare for the future interdisciplinary world of work (engineering and finance, projects), to broaden academic skills with respect to optimization and to obtain interdisciplinary competences.	
Literature & Study Materials	Principles of Corporate Finance, Brealy, Myers and Allen, Eleventh version or later, Global Edition Related academic papers	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/.	
Permitted Materials during Exam	* The use of a calculator is allowed, but no mobile phones or tablets / iPads. * No books or dictionaries are allowed	
Collegerama	Yes	

CME4000	Project Management	6
Responsible Instructor	Drs. M. Leijten	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course focuses on the project management of construction projects. Projects, in all contemporary industry sectors and specifically in construction, are becoming increasingly complex and challenging with the accelerated advances in technology, new business models, forms of collaboration, availability requirements, and an augmented demand to deliver economic value and a competitive advantage. Managing projects in this environment requires strong leadership skills and proficiency in project management knowledge and practice to be able to achieve the project and organizational objectives on time, on budget and to the satisfaction of stakeholders. The course is designed to provide students with scholarly knowledge in the practice of managing construction projects in order to enhance their career options and prepare them to move into management roles by developing their professionalism, versatility and leadership in an environment of constant change. The course presents the foundations of project management, as well as more advanced academic analyses on the matter and has been developed primarily to understand the role of project management in construction. It provides an introduction to the methods and tools of project management and how they function in the process of constructing an asset. The course provides students with the knowledge necessary to understand how projects can be managed. The curriculum follows the competencies required to move a project through its various stages. The course provides a thorough exploration of project management processes and tasks. These include: - Project organisation - Planning tasks and control strategies - Contracts and procurement methods - Communication and relationships between project stakeholders. - Risk analysis and management	
Study Goals	Intended learning outcomes of this course are: - Understand and apply tools and techniques to effectively analyse and manage construction and engineering projects. - Understand the coherence and interrelatedness of technical and human aspects in managing projects and develop strategies to manage these technical and human aspects in coherence. - Develop management strategies to effectively deliver a construction project. - Develop critical and strategic thinking on and insight in the application of project management tools and techniques. - Report effectively and with a solid argumentation on developed project management strategies. - Collaborate effectively in a team. At the end of the course students should meet these project management learning outcomes: - Understand and apply key-project management tools for estimation of cost and schedules of projects, as well as resource planning to effectively develop an optimised project development and execution plan. - Understand and apply techniques for risk analysis and assessment and risk response planning to optimise the chance of successful project execution. - Understand, evaluate and apply key-contributors to a project contracting and procurement strategy to effectively manage outsourcing. - Understand and apply a project performance monitoring technique (earned value analysis) - Understand and apply performance control indicators by developing an advice on corrective measures - Understand change management and develop an advice on a change request - Understand and apply theories on: * Project complexity * Project value * Conflict management * Project organisation * Control versus commitment	
Education Method	Live lectures and course materials offered online via the virtual learning environment, as well as group assignments. Lectures will be either on campus or online, or a mixture of the two, depending on constraints and circumstances that may be in effect.	
Assessment	A number of group assignments require a pass to complete the course. There is also an individual written exam, which is open-book. The mark for the individual written exam will be the final mark for the course, providing the student also received a pass for the group assignments. Group members can receive a bonus score of 0.5 or 1 point for the written exam in case of exceptional performance in the group assignments. This can only be applied to an already sufficient exam mark (5.8 or higher). The results of the group assignments and the written exam remain valid after the end of the academic year. A group assignment bonus will only be valid during the academic year in which the group assignments were completed. A sufficient final mark (5.8 or higher) is required to pass the course. General grading rules of Delft University of Technology apply. Exams will either be organized on campus or remotely, depending on possible constraints in effect. Each academic year, two written exam opportunities are offered.	
Elective	Yes	
Tags	Project planning / management	
Expected prior Knowledge	There are no mandatory entrance courses. Proficiency in the English language is necessary.	
Academic Skills	- Collaboration and communication in a group - Academic writing - Building an argumentation - Analysis of complex situations - Proper referencing	
Literature & Study Materials	Required literature will be announced in the virtual learning environment.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	All printed materials. Possible permission to use electronic devices will be announced in the virtual learning environment.	
Collegerama	Yes	

CME4120	Information Systems for the Construction Industry	4
Responsible Instructor	Dr.ir. G.A. van Nederveen	
Contact Hours / Week x/x/x/x	2HC, 4PT	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Information Systems for construction Management (ISM) are part of the asset and construction management processes supported by various tools, technologies involving the generation and management of digital representations of engineering assets. The alternative view within this course is to not only represent physical, and functional characteristics of the engineering asset (3D) but also represent so-called 'design for ty' information characteristics for the entire asset service life. In this course the following concepts are addressed: 1. Building information modelling and design for ty's; 2. Geometric 3D computer modelling; 3. Constructability within information systems; 4. Maintainability within information systems; 5. Safety within information systems; 6. Sustainability within information systems; 7. ISM future outlook. This course has a constructivist character, in which the student is guided and encouraged to develop open design knowledge & solutions (openings) using the self-chosen Project of Interest (PoI) as a connective vehicle enabling the Open Design Learning (ODL) response. The PoI is either being constructed or already been constructed (must have passed the initial phase)	
Study Goals	Students must be able to: 1. understand state-of-the-art information systems concepts, principles and practices for a construction project; 2. examine and relate these abstract information systems for construction concepts; 3. rework and transform the project specific information systems concepts and insights; 4. form an individual judgement on these new insights; 5. create a response that integrates all concepts, (computer) models, new insights and developed results; 6. demonstrate the internalization process of the aforementioned learning and development goals and achievements.	
Education Method	The course is based on the Open Design Learning concept (ODLc): every week students are asked to study specific concepts and apply these to a self-chosen Project of Interest (PoI) by means of a self-created response and related open-glass-box computer models. The teachers incite the ISM concepts as a reflective practitioner using both reference books and dialogue questions from the students. The students have 2 hours of these concept and dialogue sessions and 4 hours of (computer) work sessions per week. During the work sessions, students can work on their ODL response under supervision of the teacher/constructor.	
Assessment	Open Design Learning (ODL) response: an original enabler demonstrating the personal learning achievements (report/ computer model/ digital audio or video files/ animation/) The ODL response illustrates how the general concepts have been linked and evaluated to the self-chosen PoI using either a 1) logical review and/or 2) computer model.	
Expected prior Knowledge	Basic knowledge and experience of BIM, CAD or similar	
Academic Skills	Skills to: - curiously connect to a committed system of interest; - critically convince others with a compelling response; - clearly convey a communicative response; - creatively convert concepts into a comparative response; - construct a properly compiled open glass box model; - contextualize in a combined and open space; - conclude towards comprehensive open ended solutions.	
Literature & Study Materials	Reference books: - Borrmann et al, "Building Information Modeling" - Sacks et al, "BIM Handbook" Other material TBA.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE):	
Permitted Materials during Exam	n/a	
Collegerama	Yes	

CME4150	Managing Uncertainty and Data	4
Responsible Instructor	Dr.ir. M. Nogal Macho	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Ir. J.S. Hoving	
Contact Hours / Week x/x/x/x	2HC, 4PT	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Construction Managers and Engineers have to make decisions on a daily basis. In some cases, the decisions are critical and can cause a project to be successful or to fail. To ensure rational, unbiased decision-making, construction managers or engineers should be able to observe and make sense of different types of data. They should be able to identify the data they need, the most convenient manner of organizing or presenting the data so it can be better understood and the information they can safely derive from the data. The last point is paramount because any dataset is associated with a given amount of uncertainty. In other words, construction management implies making decisions in conditions of uncertainty, thus, identifying and incorporating such uncertainty into the process will result in well-grounded decisions. The course Managing Uncertainty and Data (MUD) will provide students with the knowledge and tools to adequately deal with data and uncertainty to support their decision-making process during their professional development as construction managers and engineers. The topics presented in the MUD course can be classified into data- and uncertainty-related as follows; DATA Data collection: sensing and monitoring Observation theory Model-based data generation Sensitivity analysis UNCERTAINTY Probability theory Risk and reliability Uncertainty quantification	
Study Goals	By the end of this course, the student should be able to: perform multivariate analysis of data to extract knowledge via data-driven approaches, modelling tools and uncertainty quantification methods. construct relevant probabilistic models, for example using bivariate distributions or discrete Bayesian Networks. use risk and reliability analysis concepts to evaluate systems, their characteristics and behaviour to support decision-making under uncertainty. propose actions (short-term) or plans (long-term) to address specific construction and management-related problems based on an adequate strategy for data and uncertainty management with a holistic vision of the course. accept and relate the different backgrounds and prior knowledge of team members and utilize this multidisciplinary to optimise team performance and peer learning.	
Education Method	The course learning objectives are met using the following teaching concepts: Transfer of theory and knowledge through Lectures & Discussions Engagement with theoretical concepts through Practicals & Workshops Self-study through Videos, Quizzes and Notebooks A multidisciplinary team project, including both supervisor and peer review Formative online tests to monitor learning progress This course has been designed to guarantee you can give solutions to the issues that you will face during your professional life as a construction manager or engineer. With this in mind, the course is underpinned by the following three pillars. (i) Onboarding and synergies within a multidisciplinary environment. The CME student population is a melting pot of different cultural and educational backgrounds and is therefore by default multidisciplinary. To be able to collaborate effectively within the MSc CME and specifically within this course, all CME students must (1) be aware of the existing multidisciplinary among their population, (2) understand that this multidisciplinary is a strength that can be utilized and (3) embrace this multidisciplinary as an opportunity to learn from each other. As part of this course, students work together during a multidisciplinary team project on Uncertainty & Data management. The team composition is determined by the lecturers based on student background and prior education, ensuring multidisciplinary in all project teams. Throughout the course, the project teams will participate in several group sessions and activities. To become aware of the existing multidisciplinary in their project teams, the students in each team will meet during the first week of the period to (1) meet and inform each other of their cultural and educational backgrounds, (2) identify each others strengths and weaknesses and based on this (3) organize and assign their roles and tasks in the project team. (ii) Practice-oriented course. Practice-oriented course. During the course, the students will apply the theoretical concepts to solve a realistic challenge. This will help them understand the relevance of the course in real practice, regardless of the final orientation of their professional careers. To give an answer to the challenge, they will have to prioritise information, make sound assumptions and derive conclusions supported by data, in the same manner as they will do during their professional life. The skills and tools learned during this course become essential to building up a solution to the challenge. Also, the students will realise how as they progress in the course, they will be able to better communicate with their team with more specific terminology, a better-organised data presentation and with statements that are supported by the data rather than based on biased perceptions. (iii) Online progressive self-assessment. Throughout the course, students may test their own learning progress by repeatedly taking an online test that consists of randomised problems regarding uncertainty and data management. This allows the students to monitor their knowledge, current level and points of improvement on the different course topics throughout the course period. This also allows the lecturers to provide feedback and monitor the learning of the students. At the end of the course, the students must have reached the pass threshold for the online test, where the pass/non-pass requirement is established at a team level	
Assessment	To pass the course, students must pass a team project. The students will only be able to submit the team project if each member of his/her team has reached the pass threshold for the online progressive self-assessment. The grading for the course will be based on the team project only. In the team project, the students will give answers to a given challenge. They should demonstrate their learning and development achievements, by explaining the process, decisions, and	

	strategies adopted based on the concepts learned during the course.
Expected prior Knowledge	The team project will be assessed based on the course rubric. To pass the course, the grade should be higher than or equal to 6. General understanding of civil engineering systems. BSc level mathematics (calculus, linear algebra).
Academic Skills	Critical thinking Analytical thinking Problem-solving Judgement skills Reasoning Basic computing programming
Literature & Study Materials	Reader Probability and Observation Theory Teunissen PJG, Tiberius CCJM, Verhagen S Ale, B.J.M., 2009. Risk: An Introduction: The Concepts of Risk, Danger and Chance. Available as PDF file and hardcopy on https://www.bol.com/nl/nl/f/risk/33102297/
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

CME4200	Intercultural Relations	2
Responsible Instructor	Dr.ir. C. van Daalen	
Responsible for assignments	Dr. S. Hinrichs-Krapels	
Co-responsible for assignments	Dr.ir. V.J. Cortes Arevalo	
Contact Hours / Week x/x/x/x	Two workshops	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This course introduces students to some theories on diversity and dimensions of national cultures as defined by Hofstede and others. Through role-playing games and exercises the students engage with the theory and reflect on their own intercultural experience and develop their cross-cultural sensitivity. Departing from sociological and psychological theories on communication and in-group/out-group this course will treat themes like diversity, inclusiveness and intercultural cooperation. It takes a dynamic and multifaceted approach to culture as its starting point. A TOPOI analysis and writing assignment will be used as a test at the end of the course.	
Study Goals	At the end of this course you will be able to... - point out the effects of globalization, realize and recognize the existence of different rationales, behaviours and cultures and you will be aware of cultural heterogeneity. - apply the TOPOI model to assess and address the impact of these differences on interpersonal and intergroup interactions. - discover and discuss cultural dimensions in given social/organisational phenomena and events. - demonstrate cultural awareness by developing an appropriate attitude towards cross-cultural collaboration and communication in given cultural contexts.	
Education Method	Two intensive half-day small-group training sessions, consisting of role-play and/or negotiation games (if Covid conditions allow). Blended classroom, pre-supposing self-study of the obliged and recommended materials: video clips, literature and popular articles offered in Brightspace. Online lectures Literature study and execution and reporting of TOPOI analysis. See also 'special information' below.	
Course Relations	This course lays a foundation for the CME1201 Collaborative design and engineering course; it sensitizes students for cultural differences and improves their intercultural competences.	
Books	Edwin Hoffman and Arjan Verdooren. 2018. Diversity Competence. Cultures dont meet; people do. Uitgeverij Coutinho. Isbn 978 90 469 0598 2	
Assessment	Presence and active participation in each of the two workshop is expected. Grading is based on a written test in the form of a short reflective report/essay in which the TOPOI model is applied. NB The TOPOI analysis is introduced in the course book by Hoffman and Verdooren (2018). In case of failure (grade below 5.8) students can hand-in an individual addition to the project that can be awarded a maximum grade of 6.0. Deadline for the addition is 2 weeks after the announcement of the grades. The exact instruction for the addition will be announced on Brightspace.	
Special Information	If circumstances prevent the game-play some additional online materials may be offered. All materials are organized in Brightspace in a blended learning fashion requiring and expecting the students to prepare for interactive online meetings, which will then be organized to discuss the study materials.	
Expected prior Knowledge	No prior knowledge is expected	
Academic Skills	Students are expected to write a short reflective essay; the format of which is provided.	
Literature & Study Materials	Video clips and a limited number of scientific articles will be made available through BrightSpace. Course book: Edwin Hoffman and Arjan Verdooren. 2018. Diversity Competence. Cultures dont meet; people do. Uitgeverij Coutinho.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Permitted Materials during Exam	not relevant	
Collegerama	No	

CME4300	Engineering Asset Management	5
Responsible Instructor	Prof.dr.ir. A.R.M. Wolfert	
Instructor	Ir. R. Binnekamp	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Engineering Asset Management (EAM) is a holistic approach to manage a system/portfolio of engineering assets (i.e., physical objects) that enables a certain quality of services (QoS), overarching and/or encompassing the worlds of Project & construction management and Operations & maintenance management (DBMO). EAM strategically balances and designs these QoS performance levels between: (a) economic user demands; (b) ecologic capital supply (c) isonomic organizational and financial means/support; embedded within a multi-stakeholders / multi-dimensional tripartism system context (open-source). EAM is integratively concerned with efficient and effective control and care of both one-off new engineering asset deliveries (projects management) and continuously safeguarding the desired quality of service levels of existing in-service assets (operations management) over the entire service life of the engineering assets base. These project and operations activities will be integrated into a strategic asset management planning system using multi-criteria decision/design optimization modelling (MCDO, open glass-box). Finally, this course will introduce some of the following important EAM topics (1) systems safety (2) resilience (3)outsourcing: a game of cooperation and (4) BIMSI asset management information. The first alternative view within this course is to provide a integrative viewpoint on EAM where it is viewed as a cyclic socio-eco purpose balancing act (economical, ecological, social). The second alternative viewpoint is the use of a service provider for reflectively practice the integrative projects and service management concepts into a multi-level strategic asset management plan. The third alternative viewpoint, is that the SPoIs EAM systems will be improved and or appraised using the Open Design principles: multi-criteria Open glass-box decision analysis/optimization , Open source systems theory, Open ended systems thinking combined with Open loops management. In this course the following concepts are addressed: 1. Your Service Provider of Interest (SPoI) and the Quality of Service (QoS) systems concept (introduction to strategic engineering asset -, project- and operations management and open design systems theory) 2. Zooming OUT the SPoIs embedding systems context and open-sources stakeholder constraints/ regulations 3. Zooming IN the open socio-eco purpose EAM system, tripartism system modelling and Open loops management (ODL, MI,MII) 4. Multi-criteria Quality of Service Decision Evaluation and Design optimization (MCDA/ MCDO) - The Projects Management and/or Operations Management solution space 5. Strategic Asset Management Planning (SAMP) integrating the Project & Construction Management and the Operations & Maintenance Management (the Project Delivery and the Service Operation Plan,: PDP and SOP) 6. Keeping the SPoI engineering assets Safe - Integral System Safety 7. Special EAM topics (1) Systems Resilience engineering (2)Outsourcing: a Game of Cooperation (3) the BIMSI EAM information concept For more course info see : https://www.open-design.school/ This course has a constructivist character, in which the student is guided and encouraged to develop open design knowledge & solutions (openings) using the self-chosen Service Provider of Interest (SPoI) as a connective and practicing vehicle enabling the Open Design Learning (ODL) response. The SPoI can a transportation-, water management-, energy infrastructure and/or a real-estate system service provider.	
Study Goals	Students should be able: - To understand and be familiarized with the EAM concepts, principles and practices, by dialoguing these with the constructors, by navigating through the EAM reference books and by engaging to the SPoI. - To apply and relate these modellingEAM concepts, by dialoguing and experiencing these with the SPoI and the reflective practice. - To (re)work this EAM knowledge, by transforming and linking the SPoI-dialogues into new openings/ learning insights. - To demonstrate and develop the personal outcome of the aforementioned learning goals, by internalizing the openings/ learning achievements in an original ODL response integrating all (computer) models.	
Education Method	The course is based on the Open Design Learning concept (ODLc): every week students are asked to study specific concepts and apply these to a self-chosen Service Provider of Interest (SPoI) by means of a self-created ODL response and related open-glass-box computer models. The teachers incite the EAM concepts as a reflective practitioner using both reference books and dialogue questions from the students. The students have 2 hours of these concept and dialogue sessions and 4 hours of (computer) practical work sessions per week. During the work sessions, students can work on their ODL response under supervision of the teacher/constructor. During these practicals also Master class sessions will be organized.	
Assessment	The deliverable of this course is the so-called Open Design learning response. This ODL response is a group deliverable based on the self-chosen SPoI (one group delivers one ODL response). The Open Design Learning (ODL) response is an original enabler demonstrating both the group and personal learning and development achievements. The ODL must contain a clear justification of the individual contributions of each group member. Each group member must also write an individual Comment in which he/she writes a collegial review of at least 1 concept of the ODL response for which he/she was not responsible. Good collegial Comments make use of specific ODL Commendation aspects (see next section). All of these (incl. the open glass box model) should be presented in a self-chosen format such as report/ elaborative presentation/ digital audio or video files/ animation/ website. The ODL response illustrates how the general concepts have been linked and evaluated to the self-chosen SPoI using a: 1) logical review and/or diagramming 2) computer model(s). -- The Open Design Learning commendation principle will be applied as a grading rubric for the ODL response. Both the SPoI content characteristics, and the students learning process are integrated within these commendation principles.	

	We call it commendation because when we grade your response, we start from a grade of 10 and only deduct points if aspects are missing/only partially worked out. There are 5 commendation categories: Connect/ Construct /Conclude &Conspect /Convey /Convince
Expected prior Knowledge	CE, TPM, ABE or other engineering design related BSc.
Academic Skills	Skills to: - curiously connect to a committed system of interest; - critically convince others with a compelling response; - clearly convey a communicative response; - creatively convert concepts into a comparative response; - construct a properly compiled open glass box model; - contextualize in a combined and open space; - comprehensively conclude and cogently conspect your open ended solutions.
Literature & Study Materials	Reference books Hastings (2015) Physical Asset Management Graham M. Winch (2009). Managing Construction Projects. Wiley-Blackwell. Nigel Slack (2006). Operations Management (5th edition). Prentice-Hall. Crespo Márquez (2007). The Maintenance Management Framework, Springer. Reference reader (content) Binnekamp; Heukelum; Wolfert ;Zhilyaev (2022) Reader - Open Design Systems Reference scientific articles Kammouh, O ; Nogal, M ; Binnekamp, R ; Wolfert, A.R.M. / Multi-system intervention optimization for interdependent infrastructure. In: Automation in Construction. 2021 ; Vol. 127, pp. 1-11. Zhilyaev, D. ; Binnekamp, R. ; Wolfert, A.R.M. / Best Fit for Common Purpose : A Multi-Stakeholder Design Optimization Methodology for Construction Management . In: Buildings. 2022 ; Vol. 12, No. 5.
Judgement	After handing in the ODL response your will receive your grade using the ODL commendation principle. To pass the course your ODL grade should be higher than or equal to 6. After commending your ODL response one plenary open dialogue session (max. 3 hours, somewhere between week 12 and 14 to be announced) with one of the constructors will be scheduled. Only during this session(s) we can reflect on your ODL response: bear in mind, the outcome of this reflection can result in a lower/equal/higher grade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Specialisations

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Engineering & Systems

CME4170	Construction Technology	5
Contact Hours / Week x/x/x/x	This course will start in 2023-2024	

CME4481	Systems Engineering Design	5
Responsible Instructor	Prof.dr.ir. A.R.M. Wolfert	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Systems Engineering Design (SED) is a powerful design approach to get from a semi-unstructured multi-stakeholder set of vague and messy goals and needs to a well-defined set of requirements and complementary socio-technical systems. In this course a best fit for common purpose design methodology to design for complex interconnected systems using multi-objective optimization techniques. Within this approach both the subject preference function model (human) and the physical object behavior model (engineering asset) will be integrated. In other words, to design socio-technical systems one needs a sound and integrative mathematical approach (open glass box modelling) to support participatory decision making on new engineering artefacts (open source participation and open ended system theory). It will be demonstrated that this best fit for common purpose design methodology also can be used for Project management and/or Operations management planning applications. The alternative view within this course is to combine multi-objective (non-)linear systems optimization and the theory preference function modeling in a constructivist design methodology, using mathematical simulation and modelling. This Open Design Systems approach is distinctive from pure engineering optimization methods. Within this approach the main concepts of systems design for tY-s, decision and development are linked with a self chosen System of Interest (SoI): 1. Systems theory and thinking and design as problem (dis)solving; 2. Hard engineering asset systems (technical-physical) vs. soft human management systems (social); 3. A posteriori multi-actor design decision systems; 4. A priori multi-actor design decision systems; 5. Non-linearity in mathematical models for design/decision systems; 6. Best fit for common purpose. For more course info see : https://www.open-design.school/ This course has a constructivist character, in which the student is guided and encouraged to develop open design knowledge & solutions (openings) using the self-chosen System of Interest (SoI) as a connective and practicing vehicle enabling the Open Design Learning (ODL) response. The SoI can a transportation-, water management-, energy infrastructure and/or a real-estate system.	
Study Goals	To be familiarized with and understand state-of-the art systems engineering management systems concepts, principles and practices, by (i) dialoguing these with the constructors, (ii) navigating through the system engineering management reference documents and (iii) engaging to a self-chosen real-life System of Interest (SoI). To relate and examine these abstract systems engineering management concepts, by (i) constructing SoI specific (computer) models, (ii) dialoguing with these models and (iii) experiencing these with the SoI and its reflective practice. To rework and transform the SoI specific systems engineering management concepts observations, by (i) transforming and linking the dialogues and experiences into new insights and (ii) developing improvement results where applicable. To form an individual judgement and appraise these new insights/results by means of a conspection between these and the original concepts within the specific context of the SoI. To create an original Open Design Learning (ODL) response that (i) integrates all concepts, (computer) models, new insights and developed results and (ii) demonstrates the internalization process of the aforementioned learning goals and conveys the ODL achievements.	
Education Method	The course is based on the Open Design Learning concept (ODLc): every week students are asked to study specific concepts and apply these to a self-chosen System of Interest (SoI) by means of a self-created ODL response and related open-glass-box computer models. The teachers incite the SED concepts as a reflective practitioner using both reference books and dialogue questions from the students. The students have 2 hours of these concept and dialogue sessions and 4 hours of computer practical work sessions per week. During the work sessions, students can work on their ODL response under supervision of the teacher/constructor. During these practicals also Master class sessions will be organized.	
Assessment	The deliverable of this course is the so-called Open Design learning response. This ODL response is a group deliverable based on the self-chosen SoI (one group delivers one ODL response). The Open Design Learning (ODL) response is an original enabler demonstrating both the group and personal learning and development achievements. The ODL must contain a clear justification of the individual contributions of each group member. Each group member must also write an individual Comment in which he/she writes a collegial review of at least 1 concept of the ODL response for which he/she was not responsible. Good collegial Comments make use of specific ODL Commendation aspects (see next section). All of these (incl. the open glass box model) should be presented in a self-chosen format such as report/ elaborative presentation/ digital audio or video files/ animation/ website. The ODL response illustrates how the general concepts have been linked and evaluated to the self-chosen SPoI using a: 1) logical review and/or diagramming 2) computer model(s). -- The Open Design Learning commendation principle will be applied as a grading rubric for the ODL response. Both the SPoI content characteristics, and the students learning process are integrated within these commendation principles. We call it commendation because when we grade your response, we start from a grade of 10 and only deduct points if aspects are missing/only partially worked out. There are 5 commendation categories: Connect/ Construct /Conclude & Conspect /Convey /Convince	
Expected prior Knowledge	CE, TPM, ABE or other engineering design related BSc, including (non)linear algebra calculus	
Academic Skills	Skills to: - curiously connect to a committed system of interest; - critically convince others with a compelling response; - clearly convey a communicative response; - creatively convert concepts into a comparative response; - construct a properly compiled open glass box model; - contextualize in a combined and open space; - comprehensively conclude and cogently conspect your open ended solutions.	
Literature & Study Materials	General: Blanchard, B.S. and Fabrycky, W.J., "Systems Engineering and Analysis" Print: 5th edition July 2013. Binnekamp; Heukelum; Wolfert ;Zhilyaev (2022) Reader - Open Design Systems (addendum to this reader). Zhilyaev, D. ; Binnekamp, R. ; Wolfert, A.R.M. / Best Fit for Common Purpose : A Multi-Stakeholder Design Optimization Methodology for Construction Management . In: Buildings. 2022 ; Vol. 12, No. 5. Extra:	

	Ackoff, R.A., "Ackoffs Best". Wiley, 1999. Barzilai, J. Preference Function Modelling: The Mathematical Foundations of Decision Theory in: Trends in Multiple Criteria Decision Analysis, Springer, 2010. Binnekamp, R., Preference-based Design. IOS Press, 2010. Dyme, C.L. and Little, P., Engineering Design: A Project-Based Introduction, 2nd Edition ISBN-13: 978-0471256878, 2003. Fisher, R. and Ury, W.L., Getting to Yes. Penguin, 1991. Roozenburg, N.F.M. and Eekels, J., Product Design: Fundamentals and Methods, ISBN-13: 978-0471954651, 1995. Van Gunsteren, L.A., "Quality in Design and Execution of Engineering Practice" ISBN978-1-61499-251-6 (print) 978-1-61499-252-3 (online), 2013.
Judgement	After handing in the ODL response your will receive your grade using the ODL commendation principle. To pass the course your ODL grade should be higher than or equal to 6. After commending your ODL response one plenary open dialogue session (max. 3 hours, somewhere between week 12 and 14 to be announced) with one of the constructors will be scheduled. Only during this session(s) we can reflect on your ODL response: bear in mind, the outcome of this reflection can result in a lower/equal/higher grade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

CME4701	Construction Management Systems	5
Responsible Instructor	Prof.dr.ir. A.R.M. Wolfert	
Contact Hours / Week x/x/x/x	2HC, 4PT	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Contractor Management Systems (CMS) course is about the way a contractor organizations manages one-off construction projects of infrastructure/real estate systems or delivers ongoing operations and maintenance quality of services, as outsourced contracts from services providers or other clients. Contractors strive to manage their construction projects/service operation organization effectively and efficiently, supported by state of the art management systems. The organization can be triparted into a (1) cultural/ecological purpose, (2) product-service/economic operating model and (3) the organizational/socioeconomic support sub-system. The overall contractors identity is integrated by the interplay of these sub-systems into the so-called socio-eco purpose. The different characteristics of this view can be derived from organizational theory in which the organization is modeled as a living social organism (a living biotope). The organization therefore changes over time as a learning and developing organization. In this she goes through different maturity levels: Pioneer-Expansion-Maturity and/or phases Pioneer-Differentiated-Integrated-Associative phases, including their relevant management & leadership styles. The predominant aspects of construction project management are money, organization, time, risks, information, quality, and safety: i.e., MOTRIQS. For a service operations contractor, these most important drivers are slightly modified into MOSRIQS (where the first S stands for quality of Service and the Q for process Quality). The nature of a construction project management organization is controlling its one-off project goals and deliverables, where the nature of a service operation organization is to care for the engineering assets by safeguarding their quality of services parameters (e.g. reliability, availability, maintainability, safety etc.). Managing a contractor organization means taking different decisions because both project as well as service management are dynamic process on-the-run. For this PFM (preference function modeling) combined MCDA (multi criteria decision analysis) will be used to model and evaluate important managerial decision (e.g. a supplier selection, a construction logistics scenario etc.) This course offers an alternative view on contractor management systems which integrates the management concept of double loop learning, as applied to organizational learning and development, into the management (styles) of construction projects where the contractor organization is seen as an organic living entity fit for socio-eco purpose. Finally, these management systems will be improved and or justified using the Open Design principles: Open glass-box modelling, Open source multi-criteria stakeholders, Open ended systems theory combined with Open loops management. The following concepts are being addressed following the contractors viewpoint: 1. Management Systems and the PMS - Project & Construction Management System (and OMS Operations & Maintenance Service Management system). 2. Construction project planning & mitigation control (and 3C asset QoS optimization) 3. Multi-criteria managerial decision making (MCDA & PFM) 4. Fit for purpose Quality management 5. Information systems the BIMSI concept 6. The developing contractor organization - a living enterprise for the future 7. Associative co-makers and the supply chain For more course info see : https://www.open-design.school/ This course has a constructivist character, in which the student is guided and encouraged to develop open design knowledge & solutions (openings) using the self-chosen Contractor of Interest (CoI) as a connective and practicing vehicle enabling the Open Design Learning (ODL) response. The CoI can a project or service organisation that contracts within transportation-, water management-, energy infrastructure and/or a real-estate system scopes.	
Study Goals	understand and be familiarized with CMS concepts, principles and practices, by dialoguing these with the constructors, by navigating through the CMS reference books and by engaging to the CoI. apply and relate these abstract CMS concepts, by dialoguing and experiencing these with the CoI and reflective practitioners. (re)work this CMS knowledge, by transforming and linking the CoI-dialogues into new insights. demonstrate and develop the personal outcome of the aforementioned learning goals, by internalizing the openings/ learning achievements in an original ODL response.	
Education Method	The course is based on the Open Design Learning concept (ODLc): every week students are asked to study specific concepts and apply these to a self-chosen Contractor of Interest (CoI) by means of a self-created ODL response and related open-glass-box computer models. The teachers incite the CMS concepts as a reflective practitioner using both reference books and dialogue questions from the students. The students have 2 hours of these concept and dialogue sessions and 4 hours of (computer) practical work sessions per week. During the work sessions, students can work on their ODL response under supervision of the teacher/constructor. During these practicals also Master class sessions will be organized.	
Assessment	The deliverable of this course is the so-called Open Design learning response. This ODL response is a group deliverable based on the self-chosen SPoI (one group delivers one ODL response). The Open Design Learning (ODL) response is an original enabler demonstrating both the group and personal learning and development achievements. The ODL must contain a clear justification of the individual contributions of each group member. Each group member must also write an individual Comment in which he/she writes a collegial review of at least 1 concept of the ODL response for which he/she was not responsible. Good collegial Comments make use of specific ODL Commendation aspects (see next section). All of these (incl. the open glass box model) should be presented in a self-chosen format such as report/ elaborative presentation/ digital audio or video files/ animation/ website. The ODL response illustrates how the general concepts have been linked and evaluated to the self-chosen SPoI using a: 1) logical review and/or diagramming 2) computer model(s). -- The Open Design Learning commendation principle will be applied as a grading rubric for the ODL response. Both the SPoI content characteristics, and the students learning process are integrated within these commendation principles. We call it commendation because when we grade your response, we start from a grade of 10 and only deduct points if aspects are missing/only partially worked out. There are 5 commendation categories: Connect/ Construct /Conclude &Conspect /Convey /Convince	
Expected prior Knowledge	CE, TPM, ABE or other engineering design related BSc and preferably CME 4300 engineering asset management course	

Academic Skills	Skills to: - curiously connect to a committed system of interest; - critically convince others with a compelling response; - clearly convey a communicative response; - creatively convert concepts into a comparative response; - construct a properly compiled open glass box model; - contextualize in a combined and open space; - comprehensively conclude and cogently conspect your open ended solutions.
Literature & Study Materials	Reference works are L.A. Van Gunsteren (2011) Stakeholder-oriented Project Management: Tools and Concepts F. Glasl. (1997) The Enterprise of the Future: Moral Intuition in Leadership and Organizational Development (Social Ecology) Henry Mintzberg, Structure in fives: designing effective organizations 1992 Chris Argyris and Donald Schon, Organizational Learning II, 1996 Graham Winch, Managing Construction Projects, Wiley Blackwell 2015 Wayne J. Del Pico, Project Control - Integrating Cost and Schedule in Construction, RSMears 2013 Reference papers Kammouh, Omar ; Nogal, Maria ; Binnekamp, Ruud ; Wolfert, A. R.M.Rogier. / Mitigation Controller : Adaptive Simulation Approach for Planning Control Measures in Large Construction Projects. In: Journal of Construction Engineering and Management. 2021 ; Vol. 147, No. 8. Kammouh, Omar ; Kok, M. W.A.(Maurits) ; Nogal, Maria ; Binnekamp, Ruud ; Wolfert, A. R.M.(Rogier). / MitC : Open-source software for construction project control and delay mitigation. In: SoftwareX. 2022 ; Vol. 18. Wolfert, Nederveen , Binnekamp (2022) Fit for purpose Building Information Modelling and Systems Integration (BIMSI) for Better Construction Projects Management, J. of Modern Project Management (JMPM) DOI NUMBER: 1019255/JMPM029011
Judgement	After handing in the ODL response you will receive your grade using the ODL commendation principle. To pass the course your ODL grade should be higher than or equal to 6. After commending your ODL response one plenary open dialogue session (max. 3 hours, somewhere between week 12 and 14 to be announced) with one of the constructors will be scheduled. Only during this session(s) we can reflect on your ODL response: bear in mind, the outcome of this reflection can result in a lower/equal/higher grade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

CME5030	Research and Development Methods for E&S	5
Responsible Instructor	Prof.dr.ir. A.R.M. Wolfert	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This course is intended to prepare students for their final MSc thesis project either with a research or with a development flavor (conjecture thinking vs. design thinking). The 4 types of problem (dis)solving will be discussed. The students will get experience with these two main research (hypotheses) and development (prototypes) approaches and associated typical Research and Development (R&D) methodologies and management. The contradictio in terminis of the term R&D management will be handled : organic leadership vs. bureaucratic management. The two conditions for successful socio-technological innovation (R&D), excellence and intrinsic motivation to a common purpose, constitute the basis for the classification of business identities in the strategic business concept of License Giver, License Taker, Jobber, Consultant, briefly referred to as the License Giver business concept. The concept of open idealized design learning & development connected with the intuitive thinking U-model is introduced (and its theory: open heart, open will and open mind). The process of developing from an idealized design (imagination, inspiration and intuition) towards best fit for common socio-eco purpose (integrating common ground, common sense, common interest:) will be incited. The concept of and thinking fast and slow (SI and SII) will be introduced. Different organizational development phases and/or maturity levels (emerge, develop and mature) and its innovative potential are discussed: (1) Pioneer- (birth and first expansion) (2) Differentiated- (on-going expansion and isolated maturity) (3) Integrated (connective maturity)- (4) Associative phase (purpose maturity). The cooperation/partnering evolution can be distinguished by management approaches and their particular partner features: 1) vertical alignment, 2) horizontal alignment. Finally different organizational embedding organizational systems dimensions will be evaluated: (1) governmental innovation interference (2) the freedom to excel (3) the power game (4) intercultural dimensions The general concepts of this course read as: 1. Scientific research Question vs. Engineering development Statement (4Q-model). 2. The license giver business concept (License giver/taker and Jobber/Consultant). 3. Open idealized design for common purpose and the will to lead 4. The living enterprise for the future rise and decline of innovative capabilities 5. Governmental interference & embedding innovation dimensions 6. The Game of Excellence freedom to excel 7. Special R&D organizational concepts (1) the power game (2) intercultural dimensions For more course info see : https://www.open-design.school/	
Study Goals	This course has a constructivist character, in which the student is guided and encouraged to develop open design knowledge & solutions (openings) using the self-chosen innovative Organisation of Interest (OoI) as a connective and practicing vehicle enabling the Open Design Learning (ODL) response. The OoI is preferably an organisation from the Engineering Architect & Construction industry Be familiarized with and understand Research and Development (R&D) concepts, principles and practices, by (i) dialoguing these with the constructors, (ii) navigating through the R&D reference documents and (iii) engaging to a self-chosen real-life R&D organization of interest (OoI). Form an individual judgement and appraise academic R&D output on whether this (i) is part of the body of knowledge (researcher- new scientific theory) or body of products (developer - new engineering system) and (ii) makes properly use of research questions, hypotheses, development statements, prototypes, including their respective R&D methods, within both the natural sciences and social sciences context. Examine and relate these abstract R&D concepts, by (i) constructing OoI specific models, (ii) dialoguing with these models and (iii) experiencing these with the OoI and its reflective practice. Rework and transform the OoI specific R&D concepts observations, by (i) transforming and linking the dialogues and experiences into new insights and (ii) developing improvement results where applicable. Form an individual judgement and appraise these new insights/results by means of a conspection between these and the original concepts within the specific context of the OoI. Create an original Open Design Learning (ODL) response that (i) integrates all concepts, (computer) models, new insights and developed results and (ii) demonstrates the internalization process of the aforementioned learning goals and conveys the ODL achievements. Reflection(1) & introduction: Concept sessions.	
Education Method	The course is based on the Open Design Learning concept (ODLc): every week students are asked to study specific concepts and apply these to a self-chosen inovative Organisation of Interest (OoI) by means of a self-created ODL response and related open-glass-box models (logical review/diagrams). The teachers incite the R&D concepts as a reflective practitioner using both reference books and dialogue questions from the students. The students have 2 hours of these concept and dialogue sessions and 4 hours of practical work sessions per week. During the work sessions, students can work on their ODL response under supervision of the teacher/constructor. During these practicals also Master class sessions will be organized.	
Assessment	The deliverable of this course is the so-called Open Design learning response. This ODL response is a group deliverable based on the self-chosen OoI (one group delivers one ODL response). The Open Design Learning (ODL) response is an original enabler demonstrating both the group and personal learning and development achievements. The ODL must contain a clear justification of the individual contributions of each group member. Each group member must also write an individual Comment in which he/she writes a collegial review of at least 1 concept of the ODL response for which he/she was not responsible. Good collegial Comments make use of specific ODL Commendation aspects (see next section). All of these (incl. the open glass box model) should be presented in a self-chosen format such as report/ elaborative presentation/ digital audio or video files/ animation/ website. The ODL response illustrates how the general concepts have been linked and evaluated to the self-chosen OoI using logical review and/or diagramming -- The Open Design Learning commendation principle will be applied as a grading rubric for the ODL response. Both the SPoI content characteristics, and the students learning process are integrated within these commendation principles. We call it commendation because when we grade your response, we start from a grade of 10 and only deduct points if aspects are missing/only partially worked out. There are 5 commendation categories: Connect/ Construct /Conclude &Conspect /Convey /Convince	
Expected prior Knowledge	BSc engineering	
Academic Skills	Skills to: curiously connect to a committed system of interest;	

Literature & Study Materials	critically convince others with a compelling response; clearly convey a communicative response; creatively convert concepts into a comparative response; construct a properly compiled open glass box model; contextualize in a combined and open space; comprehensively conclude and cogently conspect your open ended solutions. Binnekamp/Wolfert (2022) Reader- the 4Q R&D model Roozenburg and Eekels, Product Design: Fundamentals and Methods, Wiley, 1995. Van Gunsteren and Vlas, The Licence Giver Business Concept of Technological Innovation. Springer, 2022. C. Otto Scharmer, Theory U: Learning from the Future as It Emerges. Berrett-Koehler Publishers, 2009 Russell L. Ackoff, Idealized Design, Creating an Organizations Future. Wharton School Publishing, 2006 Friedrich Glasl. The Enterprise of the Future: Moral Intuition in Leadership and Organisational Development (Social Ecology), 1997. Part I
Judgement	After handing in the ODL response you will receive your grade using the ODL commendation principle. To pass the course your ODL grade should be higher than or equal to 6. After commending your ODL response one plenary open dialogue session (max. 3 hours, somewhere between week 12 and 14 to be announced) with one of the constructors will be scheduled. Only during this session(s) we can reflect on your ODL response: bear in mind, the outcome of this reflection can result in a lower/equal/higher grade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Must choose at least one, but may also choose multiple

CME4500	Engineering Systems Optimisation	4
Responsible Instructor	Dr.ir. M. Nogal Macho	
Instructor	Prof.dr.ir. A.R.M. Wolfert	
Instructor	Dr.ir. M. Nogal Macho	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Engineering Systems Optimization (ESO) provides future managers of engineering systems with the knowledge and tools required in their professional development to support a quantitative, model-based and multi-criteria decision-making process with an objective view of the information that supports the decision. The overall goal of the ESO course is to enable students to create solutions to optimize the design and/or management of real engineering processes and systems, making meaningful decisions with consideration of the trade-off between the level of abstraction of the mathematical model used to represent the engineering problem, limitations of the optimization methods and the trust in the obtained solution. Finally, this course enables students to apply these optimization methods and programming techniques for future model-based graduation projects In this course the following concepts are addressed: 1. The optimization concepts (near-optimal solution, Pareto optimality, NP-Hard problems, etc.); 2. Problem-solving techniques and methods applied in the pursuit of improved decision-making and efficiency (e.g., linear, mix-integer linear, and non-linear optimization, (meta)heuristics). 3. Multi-objective optimization with consideration of objective and subjective information. 4. Value of a priori and a posteriori information in the decision-making process in the presence of competing objectives. 5. Applications to engineering design and decision-related problems, such as resource allocation problems, facility location, supply chain management, project production activities, scheduling and structural design. Note: the ESO course is compulsory for CME students that follow the E&S specialization. Moreover, this course is highly recommended for students who are interested in developing their Masters thesis on solving problems of construction management and engineering using a quantitative model-based approach.	
Study Goals	The student should be able to Translate an engineering-related problem as a goal-oriented mathematical optimization model, through a 4-step process (i.e., problem, model, method, and solution). Formulate a mathematical optimization model that represents an engineering-related problem with meaningful criteria for the selection of variables, objective function, and constraints, considering the implications of the search method and optimal solution, applied to a realistic Case Study. Evaluate the quality of different exact and (meta)heuristic methods to provide a valid solution for the optimization model in different contexts (i.e., research-oriented goals versus daily-practice goals), with consideration of the trade-off between computational time and quality of the solution, applied to a realistic Case Study using a programming platform (Matlab). Develop solutions to support multiple criteria decision-making in the optimization process through multi-objective optimization. Elaborate strategies for the optimal design and/or management of real engineering systems based on a holistic perspective of the engineering system optimization (i.e., capabilities and limitations of methods and models) and the trade-off between the level of abstraction of the model and the trust in the obtained solution.	
Education Method	This course is designed to adapt the course content to the student's needs and interests rather than the other way around. With this in mind, the course is divided into two types of sessions; (i) One day a week, sessions are devoted to the theoretical aspects of the course, where the teacher will present the theoretical ESO background. Based on this initial information and the suggested reading material, the student should connect the new concepts to their previous knowledge. To further internalize the new concepts, during the second hour of this session, all, students and the teacher, will discuss and give solutions to questions related to applied theory during a Q&A session. The selected collection of questions will include students' questions that can be uploaded to Blackboard before the next session. This activity aims to guarantee students' engagement and maximize your reflection on the raised questions. The Q&A session is an interactive live session, and participation is highly recommended. (ii) The second day of each week is devoted to the practical sessions where, guided by the teacher, students working in groups have to apply the new concepts to their chosen Case Study. This is a live session where the students will receive tips, suggestions, and individual feedback from the teacher. Additionally, some of the case studies will be presented and discussed with the teacher and other students. The goal of this activity is to share ideas and challenges. In such a way, the student will benefit from other students experiences which can be very useful to develop the Final Project of the course.	
Assessment	This course is designed to build students' knowledge and skills through a continuous learning process. To guarantee an adequate learning process, a formative assessment is introduced within this course. Formative assessment implies that the student will receive regular formal feedback from the teacher, although it won't be graded. Therefore, the student will have to upload four deliverables (from weeks W2 to W6) related to the different topics discussed during the previous sessions. The student will only be able to submit the final project if 3 out of 4 deliverables have a positive pass (0/1). The course evaluation will be based on a Final Project. In this Final Project, the student will answer the optimization problem related to the chosen Case Study. The student should demonstrate the learning and development achievements, by explaining the process, decisions, and strategies adopted based on the concepts learned during the course.	
Elective	Yes	
Tags	Modelling Optimisation Programming concepts Project	
Expected prior Knowledge	Programming skills are recommendable, though non-necessary.	
Academic Skills	Critical thinking Analytical thinking Problem-solving Judgement skills Reasoning Computing programming	
Literature & Study	The following reference material will be used:	

Materials	Castillo, E., Conejo, A. J., Pedregal, P., Garcia, R., & Alguacil, N. (2011). Building and solving mathematical programming models in engineering and science (Vol. 62). John Wiley & Sons. Chang, K. H. (2014). Design theory and methods using CAD/CAE: The computer aided engineering design series. Academic Press. Optimization Toolbox. User's guide. Matlab R2020a. https://www.mathworks.com/help/optim/index.html?s_tid=CRUX_lftnav Global Optimization Toolbox. User's guide. Matlab R2020a. https://www.mathworks.com/help/gads/index.html?s_tid=CRUX_gn_documentation Dede, T., Kripka, M., Togan, V., Yepes, V., & Rao, R. V. (2019). Usage of Optimization Techniques in Civil Engineering During the Last Two Decades. Curr. Trends Civ. Struct. Eng, 2, 1-17.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

EPA1316A	Introduction to Data Science	5
Module Manager	Dr. T. Verma	
Instructor	Ir. I. Lefter	
Instructor	Dr. T. Verma	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 5	
Course Language	English	
Required for	All EPA Master Students	
Expected prior knowledge	Students should have some prior programming experience. It will be beneficial if you have dealt with a functional programming language like R or Python before. If you have never programmed before, I recommend doing a crash course in Python through Coursera or other online services before joining the class. This course is not about learning how to program. If you are not from the TPM faculty and still want to join the course, send the module manager an email with your motivation to join the course before the start of the course. Without receiving such a motivation email, we will not be able to accommodate you fully.	
Course Contents	This class will train students to gather, fuse and clean data from multiple sources. In order to gain useful insights into the reality of multiple problems, we will focus on urban ecosystems. With unprecedented growth in cities, policymakers are striving to find a balance between providing equal opportunity and benefits to its citizens and sustainable development. Data science will help us understand and estimate alternative implications of solutions and communicate results to a wide audience effectively. The course is divided into 5 major modules, each focusing on crucial steps in the lifecycle of a data science project. Obtain : Obtaining data from multiple [open] data sources. Scrub : Data cleaning, munging, sampling in order to consolidate all information into a dataset that is manageable, informative and relates to your problem. Explore : Exploratory data analysis to make sense of what your data is trying to say. Model : Estimation and modelling based on statistical tools such as regression, classification and clustering. Interpret : Communicating results and reflections through visualisation, storytelling and interpretable summaries. Students will need their laptops only during labs and for homework. All classroom sessions will be a mixture of interactive lectures, discussions and labs. Labs will cater to a series of exercises that will reinforce the skills and topics being presented in class. A final project will allow students to run through the entire cycle of 5 modules by obtaining and synthesising data to explore a problem of choice and interpreting and communicating their results through an article. For students who have had statistical, math or computer programming courses in their bachelors or elsewhere, this course will add to your skills by providing you with tools to become future policy-makers, consultants, data scientists, and in general, supporters of open science. The course meets for 8 hours a week for 8 weeks divided into 2 interactive and blended lecture hours, 1 discussion hour, 4 lab hours and an extra wildcard hour every week. Although attendance for all lectures is not mandatory, substantial self-study will be required to keep up with the course if you do not attend all 8 hours a week.	
Study Goals	After successful completion of this course, you will be able to: interpret and discuss data sources that are usable and relatable for a problem presented. manipulate data and consolidate all information into a dataset that is manageable, informative and relates to your problem. describe and analyse the consolidated dataset(s) to support your problem with evidence. apply models using statistical and machine learning to infer results in the process of turning data into valuable information. report results and reflections through visualisation, storytelling and interpretable summaries, especially when faced with a new dataset. and hopefully get great data-driven policy jobs in the future, or go on adventurous travels with an open mind.	
Education Method	Out of class education methods involve 1. Reading materials and watching videos assigned before lectures. 2. Solving homework/notebooks in Python to catch up with programming (ungraded, and useful to stay on track). In class education methods involve 1. Interactive lectures and discussions 2. Labs / open hours to solve homework/assignments The course will run either in person or in an online format (given the current circumstances), but not hybrid. Recordings from 2021 are available in case students miss a lecture.	
Computer Use	The course is set in Python programming language. Students will be expected to use Python notebooks on their personal laptops. If this is not possible for you for whatever reason, please contact me before the course and we will figure try to figure an alternative solution while maintaining your privacy.	
Assessment	~50% Individual assessment methods include Assignments Discussions Participation ~ 50% Group assessment methods include Final project Note: the composition of grades can shift depending on the load of individual and group work. It is mandatory to score ≥ 5.8 in both parts to pass the course. No exam	

EPA1352	Advanced Simulation	5
Module Manager	Y. Huang	
Instructor	Prof.dr.ir. A. Verbraeck	
Instructor	Y. Huang	
Co-responsible for assignments	Prof.dr.ir. A. Verbraeck	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	SEN9110 - Simulation Masterclass (elective) can be followed after this course.	
Expected prior knowledge	Basic knowledge about modeling and simulation, data analysis, probability and statistics. Basic programming skills in Python.	
Course Contents	The objective of this course is to provide its participants with a selection of topics that are state-of-the-art in simulation. The course will focus on the theoretical foundation of (potentially large-scale) data-driven simulation, as well as the corresponding methods and techniques. The course provides a set of lectures, in-class discussions in combination with lab exercises and group assignments. The participants are required to individually prepare the classes with reading assignments according to the announcements on Brightspace. In parallel, groups of students carry out lab exercises for the topics that have been studied. The lab exercises will be discussed with the teachers and marked. Key topics covered in the course include: Data quality; Data-driven modelling and simulation; Component-based modelling.	
Study Goals	After successfully completing the course, the participants will be able to: - Describe important topics in advanced simulation and explain their implications in practice; - Explain the theoretical foundation of advanced simulation methods and their abilities; - Apply a set of advanced simulation methods to case assignments for experimentation and analysis; - Reflect on the strengths and limitations of the methods; - Hypothesize on possible future developments in the simulation discipline.	
Education Method	A mix of lectures, in-class discussions, pre-class reading assignments, supervised lab exercise sessions, and hand-in group assignments. Groups are not to exceed four members.	
Computer Use	Supervised lab exercise sessions where students will work with assignments on their own computers. The tools that are needed for the course will be made available.	
Course Relations	This course is an excellent preparation for SEN9110 - Simulation Masterclass. Graduation project using advanced simulation can benefit from this course.	
Literature and Study Materials	No textbook is required. A set of scientific papers and/or book chapters will be made available through Brightspace.	
Assessment	The grading for this course is based on the result of group assignments (50%) and on a final open-book written exam (50%). Both parts must be completed with a passing grade (at least 5.75) in order to pass the course. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as a timed take-home exam (open book). The exam will be checked for plagiarism and may be discussed and assessed in an oral debriefing.	
Permitted Materials during Tests	The course materials (papers, slides, notes, books) are permitted during the written exam.	

TPM024A	Methods for Risk Analysis and Management	5
Module Manager	Prof.dr.ir. G.L.L.M.E. Reniers	
Instructor	Dr. M. Yang	
Responsible for assignments	Prof.dr.ir. G.L.L.M.E. Reniers	
Co-responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course consists of three blocks following a risk analytical approach: (A) introduction, (B) analysis, (C) assessment, treatment and follow-up. In block A Students will get familiar with the concepts and definition used in safety science and the basic steps of a risk analysis. In block B the course will focus on identification of hazards and qualitative/semi quantitative risk analysis. A selection of methods is explored to get acquainted with its application and limitations. For instance: Fault tree analysis, Event tree analysis, Bowtie diagram, Reliability Block Diagram, Markov Chain Analysis, Bayesian networks, Decision making under uncertainty. In block C the assessment of the outcomes of a risk analysis are put central. This includes acceptance, stakeholder perspectives, decision making and monitoring.	
Study Goals	After taking this course the student will have knowledge about: * the basic concepts of safety science and risk analysis * risk acceptance and decision making about risks After following this course, the student will be able to: * apply structured methods for risk identification and risk analysis	
Education Method	The following methods may be used: Lectures, exercises and workgroup assignments	
Literature and Study Materials	The following textbook is mandatory: Engineering Risk Management (Meyer and Reniers, 2016; De Gruyter Publishing) Other required and additional material will be announced via Brightspace	
Assessment	The following assessment methods may be used according to the need: written examination and/or take-home exam.	
Elective	Yes	
Category	MSc level	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Projects & People

CME2201	Dynamic Control of Projects	4
Responsible Instructor	M.G.C. Bosch-Rekveldt	
Instructor	Prof.dr. H.L.M. Bakker	
Instructor	Prof.dr.ir. M.J.C.M. Hertogh	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	This course is obligatory for CME students in the Projects & People specialization of CME (but other CME students can follow the course as an elective).	
Course Contents	Sketching the context: Each project (green field, brown field) is an intervention in an existing situation. The aim is to incorporate a system into an existing environment. This environment is increasingly complex, hence influencing the project. A project is a specific and unique development, for a unique client, on a unique location, with a unique goal, to solve a unique problem, with a unique problem solver, etc. A project needs time for the development with a number of subsequent phases: design, engineering, implementation, etc. In most cases: (1) the state of the environment (politics, stakeholders, regulations, etc.) at the start of a project is totally different from the state of the environment during the project, at the end of the project and certainly during the operation phase during the lifetime of the system, (2) the information and knowledge about the desired system in its environment increases dramatically during the project, which leads to different views and options and (3) the state of the technology to be applied in a project changes during a project. As a consequence, we often develop systems with the ideas of yesterday and with today's technology for tomorrow's people. And that's where dynamic control comes into place: we need to be able to act dynamically: the only certainty we have is that uncertainty will hit our projects! To create value, we should not only work dynamically in an adaptive way with continuous change, but also take care that the systems we deliver are adaptable to changing circumstances. Inevitably, the traditional project management should realize that the construction world will also be a part of the new short-cyclic and circular economy. We need to play with complexity; allow for more dynamic approaches in project management and to strive for solutions that maximize value.	
Study Goals	After this course you are able to: 1. Explain the dynamic character of a project, a product and a process, 2. Perform a complexity assessment on a real life complex project, 3. Recognise the relations between complex project systems and higher scale (context) systems, 4. Apply methods to deal with the effectiveness and efficiency of projects in a fast changing world, 5. Choose management approaches to deal with different types of project complexity (fit for purpose), 6. Describe the advantages and disadvantages of innovative ways for collaboration and contract types in complex environments.	
Education Method	The concept of "dynamic control" will be elaborated in two lectures a week. Part of the course has an online character (4 lectures). The dynamic practice of projects is illustrated by 4 guest lectures (45 minutes each). Students will write an essay (in small groups) and actively provide peer feedback to other teams in the development of the essay.	
Assessment	The assessment consists of two parts: an essay and 2 individual online assignments. Essays: to be written in small groups (2/3 students), in 6 submission steps: theme - structure literature draft for peer review draft to discuss with staff - final, see lecture overview. Online: 2 individual assignments, testing the knowledge on concepts of complexity & management strategy. Three partial grades will be given: Essay: 2.5 ECTS Online 1, TOE assignment: 0.5 ECTS Online 2, Management strategy: 1 ECTS Minimum partial grades should be 5, total grade to pass the course should be >5.75. Submissions will be checked for plagiarism. -- The final grade consists of the essay grade and an individual grade for the 2 online assignments. Students have to write an essay. Both partial grades should be 5 or higher and the final grade should be a 5.75 (5.8) or higher. All assignments of the course in Brightspace (ungraded as such) should be completed before the course grade can be obtained.	
Expected prior Knowledge	Given the reflective character of this course, you should have successfully completed the CME project and process management courses. Contact the lecturer if this is not the case.	
Academic Skills	Your academic writing skills will be developed by the writing of the essay. Further, reflection skills are essential for successfully completing this course.	
Literature & Study Materials	This course uses the book "Projects & People, Mastering Success", edited by Hans Bakker and Jaap de Kleijn. ISBN 978-90-812162-9-6. Next to the book, academic papers are used (through Brightspace). The MOOC MEP201 on onlinecourses.tudelft.nl is part of this course CME2201.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Permitted Materials during Exam	n/a	
Colleggerama	Yes	

CME4100	Process Management	5
Responsible Instructor	Drs. M. Leijten	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Complex construction- and engineering-related processes require collaboration between a variety of actors within a complex and dynamic environment. An environment that is influenced by various social, political and technical factors. These actors may be related to a process in various ways, e.g. as decision-makers, regulators, risk-taking participants and stakeholders, with diverging interests, resources and attitudes towards the process, being affected by a process and influencing it in different ways and degrees. In addition to the widely used project management, process management is needed to understand and adequately deal with these characteristics. This course aims at providing students with insights, concepts and skills needed to understand the nature of interaction in a multi-actor system regarding the initiation and development of, and decision-making in uncertain and dynamic situations. Students will analyse, develop and apply strategies, tools and arrangements that are part of the process management approach. There will be special attention to transitions and adaptive systems in an infrastructure and construction environment, but the principles are more broadly applicable.	
Study Goals	After successfully completing this course this course, students should: - Be able to identify and analyse the perceptions, interests, strategies and institutional conditions that shape and constrain the behaviour of stakeholders. - Understand the dynamics of multi-actor interaction processes in uncertain and highly politicised contexts. - Understand information exchange in complex processes. - Be able to cope with substantive and strategic uncertainties and to show this in description and application. - Have developed knowledge, skills and competences on how to manage divergent and conflicting interests of stakeholders and the ability to apply this in practice. - Have developed competences on arriving at broadly-shared solutions. - Understand the essentials of decision-making in multi-actor networks. - Understand and apply theories on transition management and adaptive systems. - Have acquired knowledge, skills and competences on arrangements and tools which support collaboration, and be able to apply them.	
Education Method	Teaching is done through live lectures, as well as learning materials offered via the online learning environment. There is also a group assignment that is offered in the virtual learning environment. Lectures will be offered either on campus or in the virtual learning environment, depending on possible constraints in effect.	
Assessment	A group assignment (pass/fail) and an individual written exam, which is open-book. There will be two exam opportunities every academic year. The mark for the individual written exam will become the final mark for the course, providing the candidate has received a "pass" for the group assignment. Both scores will remain valid after the end of the academic year. Students can also receive a bonus of 0.5 or 1 point for the exam in case of exceptional performance in the group assignment. This bonus will only be valid in the year of completion of the group assignment and can only be applied to an already sufficient exam mark (5.8 or higher). A sufficient final mark (5.8 or higher) is required to pass the course. General grading rules of Delft University of Technology apply. The exams will be organised either on campus or remotely, depending on possible constraints in effect.	
Elective	Yes	
Tags	Process	
Expected prior Knowledge	There are no mandatory entrance courses. Proficiency in the English language is necessary.	
Academic Skills	- Collaboration and communication in a group - Academic writing - Building an argumentation - Analysis of complex situations - Proper referencing	
Literature & Study Materials	Required literature will be announced in the virtual learning environment.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	All printed materials. Possible permission to use electronic devices will be announced in the virtual learning environment.	
Collegerama	Yes	

CME4600	Leadership and Strategic Management	4
Responsible Instructor	Prof.dr. P.W.C. Chan	
Instructor	M.G.C. Bosch-Rekvelde	
Instructor	Drs. M. Leijten	
Instructor	Prof.dr. P.W.C. Chan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The climate emergency, rapid urbanisation and resultant need to provide more affordable housing quickly, along with the urgency to transform our food, water, energy and waste infrastructure are key societal and sustainability transitions of our time. Furthermore, technological advancements in the digital space and the changing demographics are also demanding a rethink in the ways organisations function. Moreover, successive crises such as the Global Financial Crisis in the late 2000s and the more recent Covid-19 Global Pandemic have triggered calls for radical change. Put another way, if we keep doing what we have always done, then we will always get the (unsustainable) results that we have always gotten. Thus, there is a need to do things differently, and this is where leadership and strategic management can play a significant role. In tackling the societal and sustainability transitions highlighted above, simply relying on technical knowledge is not enough. These transition challenges are wicked problems, where simple solutions and incremental change are not sufficient. To address these transitions, there is also a need to inspire people from different backgrounds to activate radical change in our actions. Leadership skills are therefore critical to influence others to take the path less travelled. In addition, projects are increasingly considered and commissioned as drivers of change. Therefore, leadership is deeply connected to the delivery of projects. The key topics of this course include: a. Context: Understanding the context of leadership and strategy in transitions; b. Leadership theories: Complementing management and leadership, and shifting the mindset from planning and control (outputs) to influencing change and transformation (outcomes); c. Strategic management: Shifting the framing from strategy formulation and (long-term) planning to purpose-driven implementation and strategy-as-practice; d. Creativity and innovation: Shifting towards more open forms of innovation, and the role leadership and organisational strategy can play; e. Communication and negotiation: Understanding the power of communication and negotiation practices to influence outcomes; f. Values and impacts: Critically reflecting on the value, values and impacts of leadership and strategic management.	
Study Goals	The aim of this course is to enable students to develop their leadership skills and knowledge about strategic management of engineering projects. The intended learning outcomes include: (a) to develop critical awareness of leadership theories and apply these appropriately in analysing their personal leadership preferences; (b) to explain the significance and process of leadership in influencing others to achieve beneficial outcomes in the management of projects; (c) to develop critical awareness and knowledge of the strategic role that project managers and project management play in supporting wider organisational change outside of the boundaries of the project organisation, and; (d) to develop knowledge of the management of projects as a value creating process that translates policies and strategies into practice.	
Education Method	Lectures, Discussions, Group and Individual Assignments	
Assessment	It is important that the teaching, learning and assessment strategy takes into account that leadership development requires the individual to critically reflect in practice. The students will be required to submit two written assignments, including (1) a group assignment that combines leadership and strategic management, and; (2) a critical review and reflection of personal leadership as applied to the management of projects.	
Expected prior Knowledge	Not applicable	
Academic Skills	Critical Thinking, Reflective Practice, Communication (Written and Oral Presentation), and Working in Teams	
Literature & Study Materials	There will be directed reading provided each week to cover different aspects of leadership and strategy management in relation to managing projects. The course is also connected with the urgent need to drive societal and sustainability transitions. The following report commissioned by the Association for Project Management provides the contextual background for this course. Morris, P. W. G. (2017) Climate Change And What The Project Management Profession Should Be Doing About It: A UK Perspective, Princes Risborough: Association for Project Management, https://www.apm.org.uk/media/7496/climate-change-report.pdf.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Permitted Materials during Exam	Not applicable	
Collegerama	No	

CME5020	Research Methodology for P&P/D&I	3
Responsible Instructor	J. Ninan	
Instructor	Dr. M.L.C. de Bruijne	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This research methodology course is intended for students in the CME Projects and People (P&P) and Design and Integration (D&I) specializations. The topics that will be presented in this course are: Defining research objectives, setting up a research framework, formulating research questions, selecting the fitting research methods along with associated data collection and analysis, planning your research, ethics in research, reporting and presenting your research. A broad palette of research methods will be introduced and discussed, including research methods such as literature review, case studies, mixed methods, qualitative comparative analysis, Q-methodology, etcetera.	
Study Goals	To be able to clearly describe a research goal for the CME M.Sc. thesis, To be able to design and conduct a scientific research, To critically evaluate and reflect on the outcome of a scientific research	
Education Method	The course will be delivered through weekly lectures and workshops. The lectures and workshops will be delivered by experts on each method.	
Assessment	The assessment of the course will be done via weekly group assignments and a final individual assignment.	
Expected prior Knowledge	No prior knowledge required.	
Academic Skills	The students will develop various skills for academic research such as conducting a literature review, and critical thinking guided by various methods, along with a focus on the research process.	
Literature & Study Materials	Lecture materials will be uploaded in Brightspace	
Judgement	The evaluation will be conducted by course instructors.	
Permitted Materials during Exam	There will not be any final exam. Evaluations will be through weekly group assignments and a final individual assignment.	
Collegerama	No	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Must choose at least one, but may also choose multiple

CIE5981	Forms of Collaboration in Civil Engineering	4
Responsible Instructor	Ir. L.P.I.M. Hombergen	
Instructor	Ir. L.P.I.M. Hombergen	
Instructor	Prof.dr.ir. M.J.C.M. Hertogh	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In course CIE5981 a review is given of the most common forms of collaboration in realising a project in civil engineering. The course discusses the state of the art. This is done in a fundamental way but the theory is clarified by means of practical examples illustrated by visiting lecturers. The course aims at preparing students fundamentally for the various forms of collaboration they will engage during his professional career. However it is emphasized that little attention will be paid to the literal contents of the various contracts. It is a matter of insight so that later on the correct choices can be made for the adequate form of contract for a specific type of project. The course is a synthesis of aspects in procurement and contracting which are partly also given in other courses. The course is both an introduction to the business of building projects for more technically oriented students as a course for more building management oriented students with practical references. The following subjects will be dealt with: principles of an agreement and the elements that play a role in collaboration are discussed the control of a project in relation to collaboration forms the contract and the corresponding components such as tasks, responsibilities and authorities the systems of reimbursement as a function of contract form risks, risk distribution, risk management, in various contract forms the family of: design & construct, DBM, DBMOT, DBMFOT, partnering, alliances, public private partnership practical examples illustrated by visiting lecturers foreign forms of collaboration the selection and choice of a contract partner new development in different countries.	
Study Goals	The students knowledge and skill in the following activities will be increased relative to the intellectual development level attained during his or her BSc study: - The understanding of the principles of project agreements; - The understanding and choice of forms of collaboration; - The evaluation of alternative forms of contract; - The various types of reimbursement; - The selection and choice of contract partners; - The understanding of the various components of contracts; - Foreign contracts .	
Education Method	lectures	
Literature and Study Materials	obligatory lecturenote(s)/textbook(s): Reader: "Forms of collaboration in civil engineering" Available at BookShop Civil Engineering. recommended other materials: Material and presentations made available on BrightSpace	
Assessment	A written, eventually Multiple Choice, exam together with an essay. Insight and overview will be tested to assess real life project situations. -- Exam and essay	
Remarks	Summary Parties in the building industry can choose between various forms of collaboration and contract models. The different forms of collaborations are derived from the economic concepts such as value, price and cost. The course has a theoretical character but its contents is illustrated by means of practical examples by guest lecturers. The following forms of collaboration are dealt with: traditional contract, building team, general contracting and the families of Design & Construct. Discussed will be the contract-content with tasks, obligations, authorities, responsibilities, liabilities, systems of reimbursement and risk division. The following models of contract are discussed: design & construct, partnering, alliances, 2-phase contracts, public private partnership, risk management, risk sharing and contracts that deal with the organization during building.	
Expected prior Knowledge	General academic knowledge	
Academic Skills	Ability to analyse some different business concepts	
Literature & Study Materials	Reader, information on Brightspace, including some scientific papers	

Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Dictionary
Collegerama	No

CME4800	Procurement of Complex Public Projects	5
Responsible Instructor	Dr.ir. A. Straub	
Responsible Instructor	Ir. J.H. Veldhuis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course provides an overview of the basic principles of public procurement in construction. By explaining and exploring different project delivery models and related procurement methods and tender evaluation models, insights are given in the choices of the public client when designing a tender and the impact of these choices on the client organization and the client suppliers relation. Based on the case of a complex project, all elements of a procurement process in its context (characteristics project, public values, stakeholders, supply market, responsibilities, activities and risks, and the life-cycle perspective) is discussed and a procurement strategy for a complex project is designed.	
Study Goals	At the end of this course you should be able to: LO1: explain the role of public clients and public values in construction; LO2: analyse the context of procuring complex projects; LO3: critically reflect on the application of project delivery methods; LO4: design a procurement strategy for complex projects; LO5: writing a scientifically based advisory report using provided literature.	
Education Method	The course includes theoretical lectures, guest lectures and group tutorials. The course is structured around a real case upon with the assigned tasks are based.	
Assessment	Presentations and written assignments.	
Expected prior Knowledge	Interest in public clients and values, and procurement strategies for complex projects in the built environment.	
Academic Skills	Students are expected to critically reflect on lectures and provided literature and to apply this to a case by writing a scientific advisory report. Students have to collaborate with other students to deliver a group assignment.	
Literature & Study Materials	Case documents, video's and scientific literature are made available via Brightspace.	
Judgement	Evaluation is based on a group assignment (30%) and an individual assignment (70%). Both assignments have to be graded minimum 5,8 to pass the course.	
Permitted Materials during Exam	Not Applicable	
Collegerama	No	

TPM024A	Methods for Risk Analysis and Management	5
Module Manager	Prof.dr.ir. G.L.L.M.E. Reniers	
Instructor	Dr. M. Yang	
Responsible for assignments	Prof.dr.ir. G.L.L.M.E. Reniers	
Co-responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course consists of three blocks following a risk analytical approach: (A) introduction, (B) analysis, (C) assessment, treatment and follow-up. In block A Students will get familiar with the concepts and definition used in safety science and the basic steps of a risk analysis. In block B the course will focus on identification of hazards and qualitative/semi quantitative risk analysis. A selection of methods is explored to get acquainted with its application and limitations. For instance: Fault tree analysis, Event tree analysis, Bowtie diagram, Reliability Block Diagram, Markov Chain Analysis, Bayesian networks, Decision making under uncertainty. In block C the assessment of the outcomes of a risk analysis are put central. This includes acceptance, stakeholder perspectives, decision making and monitoring.	
Study Goals	After taking this course the student will have knowledge about: * the basic concepts of safety science and risk analysis * risk acceptance and decision making about risks After following this course, the student will be able to: * apply structured methods for risk identification and risk analysis	
Education Method	The following methods may be used: Lectures, exercises and workgroup assignments	
Literature and Study Materials	The following textbook is mandatory: Engineering Risk Management (Meyer and Reniers, 2016; De Gruyter Publishing) Other required and additional material will be announced via Brightspace	
Assessment	The following assessment methods may be used according to the need: written examination and/or take-home exam.	
Elective	Yes	
Category	MSc level	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Design & Integration

CME4100	Process Management	5
Responsible Instructor	Drs. M. Leijten	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Complex construction- and engineering-related processes require collaboration between a variety of actors within a complex and dynamic environment. An environment that is influenced by various social, political and technical factors. These actors may be related to a process in various ways, e.g. as decision-makers, regulators, risk-taking participants and stakeholders, with diverging interests, resources and attitudes towards the process, being affected by a process and influencing it in different ways and degrees. In addition to the widely used project management, process management is needed to understand and adequately deal with these characteristics. This course aims at providing students with insights, concepts and skills needed to understand the nature of interaction in a multi-actor system regarding the initiation and development of, and decision-making in uncertain and dynamic situations. Students will analyse, develop and apply strategies, tools and arrangements that are part of the process management approach. There will be special attention to transitions and adaptive systems in an infrastructure and construction environment, but the principles are more broadly applicable.	
Study Goals	After successfully completing this course this course, students should: - Be able to identify and analyse the perceptions, interests, strategies and institutional conditions that shape and constrain the behaviour of stakeholders. - Understand the dynamics of multi-actor interaction processes in uncertain and highly politicised contexts. - Understand information exchange in complex processes. - Be able to cope with substantive and strategic uncertainties and to show this in description and application. - Have developed knowledge, skills and competences on how to manage divergent and conflicting interests of stakeholders and the ability to apply this in practice. - Have developed competences on arriving at broadly-shared solutions. - Understand the essentials of decision-making in multi-actor networks. - Understand and apply theories on transition management and adaptive systems. - Have acquired knowledge, skills and competences on arrangements and tools which support collaboration, and be able to apply them.	
Education Method	Teaching is done through live lectures, as well as learning materials offered via the online learning environment. There is also a group assignment that is offered in the virtual learning environment. Lectures will be offered either on campus or in the virtual learning environment, depending on possible constraints in effect.	
Assessment	A group assignment (pass/fail) and an individual written exam, which is open-book. There will be two exam opportunities every academic year. The mark for the individual written exam will become the final mark for the course, providing the candidate has received a "pass" for the group assignment. Both scores will remain valid after the end of the academic year. Students can also receive a bonus of 0.5 or 1 point for the exam in case of exceptional performance in the group assignment. This bonus will only be valid in the year of completion of the group assignment and can only be applied to an already sufficient exam mark (5.8 or higher). A sufficient final mark (5.8 or higher) is required to pass the course. General grading rules of Delft University of Technology apply. The exams will be organised either on campus or remotely, depending on possible constraints in effect.	
Elective	Yes	
Tags	Process	
Expected prior Knowledge	There are no mandatory entrance courses. Proficiency in the English language is necessary.	
Academic Skills	- Collaboration and communication in a group - Academic writing - Building an argumentation - Analysis of complex situations - Proper referencing	
Literature & Study Materials	Required literature will be announced in the virtual learning environment.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	All printed materials. Possible permission to use electronic devices will be announced in the virtual learning environment.	
Collegerama	Yes	

CME4400	Entrepreneurial Engineering	5
Responsible Instructor	Dr. D.F.J. Schraven	
Responsible for assignments	Ir. L.P.I.M. Hombergen	
Co-responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Co-responsible for assignments	Prof.dr.ir. M.J.C.M. Hertogh	
Co-responsible for assignments	Prof.dr.ir. F.E.H.M. Smulders	
Contact Hours / Week x/x/x/x	This course starts in 2021-2022	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	CME students, the specialisation Design & Integration.	
Course Contents	The course deals with the entrepreneurial traits in the civil engineering domain. Students are introduced with innovation and sustainability transition theories, new economy concepts and business concepts. They are challenged by 2 assignments how to implement a technological innovation in the civil engineering market structure and processes. The course consists of two cumulative modules: Analysis module, in which you analyse the success of a retrospective innovation case as a group, informed by innovation and transition theories and tools as your basis and inspired by relevant guest lectures from innovation experts from the civil engineering practice. Development module, in which you create a business plan for a real-time technological innovation as a group, informed by business model innovation processes, transition and adaptation methods.	
Study Goals	Finishing the course, that students are able to: - Understand and analyse an innovation case, with innovation theory to identify the successes and failures on its adoption; - Understand specific market structures of the civil engineering domain, the relations between public and private roles in this domain, and the multifaceted reality thereof; - Understand the market processes, specifically the public tendering processes for innovation-oriented projects; - Analyse the technical aspects of a technology / innovation to identify the robustness and the alternative implementations thereof; - Analyse the operational context of the projected implementations of a technology; - Create the role of your business in the pursuit of implementing the technology / innovation; - Create an adaptive and embedding business plan (flexible and future oriented) for the case. - From the perspective of a client or an established market party: be able to identify, stimulate, guide and embrace innovative innovation concepts.	
Education Method	The course is an activity-based learning experience, which is accompanied by lectures with the intent to inform and inspire the student with theoretical and practical insights.	
Assessment	Both in the Analysis and Development module, the assignment will be prepared in groups in the form of a report.	
Expected prior Knowledge	No prior knowledge is required.	
Academic Skills	To critically analyse the success of a previous innovation and extract lessons from it, and to be able to develop a business plan and model for an innovation in a civil engineering context.	
Literature & Study Materials	A selection of Book Chapters, academic articles and case-based data.	
Judgement	The assignments make up 50% each of the final grade. The combined grade of the 2 group assignments should be a 6.0 or higher in order to pass. Also both assignments should minimally be a 5.0 to count.	
Permitted Materials during Exam	N/A	
Collegerama	No	

CME5020	Research Methodology for P&P/D&I	3
Responsible Instructor	J. Ninan	
Instructor	Dr. M.L.C. de Bruijne	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This research methodology course is intended for students in the CME Projects and People (P&P) and Design and Integration (D&I) specializations. The topics that will be presented in this course are: Defining research objectives, setting up a research framework, formulating research questions, selecting the fitting research methods along with associated data collection and analysis, planning your research, ethics in research, reporting and presenting your research. A broad palette of research methods will be introduced and discussed, including research methods such as literature review, case studies, mixed methods, qualitative comparative analysis, Q-methodology, etcetera.	
Study Goals	To be able to clearly describe a research goal for the CME M.Sc. thesis, To be able to design and conduct a scientific research, To critically evaluate and reflect on the outcome of a scientific research	
Education Method	The course will be delivered through weekly lectures and workshops. The lectures and workshops will be delivered by experts on each method.	
Assessment	The assessment of the course will be done via weekly group assignments and a final individual assignment.	
Expected prior Knowledge	No prior knowledge required.	
Academic Skills	The students will develop various skills for academic research such as conducting a literature review, and critical thinking guided by various methods, along with a focus on the research process.	
Literature & Study Materials	Lecture materials will be uploaded in Brightspace	
Judgement	The evaluation will be conducted by course instructors.	
Permitted Materials during Exam	There will not be any final exam. Evaluations will be through weekly group assignments and a final individual assignment.	
Collegerama	No	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Must choose one, but may also choose multiple

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	

AR2MBE025	Urban and Infrastructure (Re)development Game	10
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Y. Chen	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	

TPM023A	Cost-Benefit Analysis: Theory and Application	4
Module Manager	Dr.ir. Z. Roosenboom-Kwee	
Instructor	N. Mouter	
Responsible for assignments	Dr.ir. Z. Roosenboom-Kwee	
Co-responsible for assignments	N. Mouter	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Summary	The course TPM023a focuses not only on private (e.g. project, firm, industry) costs and benefits but also on social costs and benefits. While the private CBA only considers financial aspects, the social CBA considers both financial and non-financial aspects. For instance, in the construction of new highway, besides the financial aspects such as toll revenues or construction costs, we also assess the non-financial aspects such as safety benefits (lives saved, accidents avoided). Hence, the course actually has a much broader view than the private CBA which offers students a higher level of practical applications.	
Course Contents	Cost-benefit analysis (CBA) has become a widely used (often mandatory) tool for public policy decision making on infrastructural projects or environmental and climate policies. Through CBA, the potential current and future gains and losses of a proposed policy are identified, and then converted into monetary units in order to make them comparable with policy alternatives; certain decision rules help determine whether the project is desirable from society's standpoint. Because CBA evaluates policy and/or project proposals from the point of view of the public interest, the market prices used in CBA are corrected for possible market distortions or failure (for example, un-priced environmental damage). Nevertheless, the application of CBA is fraught with difficulties and limitations that need to be taken into account. What rates of interest are appropriate for determination of present and future costs and benefits of a policy proposal? Why and how do the private benefits and costs of a policy proposal differ from the social benefits and costs? How can we express e.g. environmental damage, or human health and life risks, in monetary units? How can we adjust market prices to account for external effects? How can we use alternative measures of welfare to identify the benefits and costs of a policy proposal? How can we ensure that politicians who aspire to use the results of CBA in their decisions can actually do so?	
Study Goals	At the end of this course, students will be able to: * understand why CBA can be used to appraise projects with an objective to improve social welfare * gain an understanding of the fundamentals of CBA * perform steps or approaches to evaluate the financial and economic values of a project * understand the strength and limitations of CBA in project appraisal. * interpret the results of a cost-benefit analysis * criticize and defend the CBA methodology from an ethical perspective * explain the advantages and the disadvantages of applying a CBA methodology in a political process with multi-stakeholders	
Education Method	The lectures are on campus. However, depending on the COVID-19 measures, the lectures may be given through a hybrid form or online and remote form.	
Literature and Study Materials	* Textbook: Anthony E. Boardman, David H. Greenberg, Aidan R. Vining & David L. Weimer: Cost-Benefit Analysis: Concepts and Practice, 4th International Edition, Pearson, 2013. * Selected articles (available in Brightspace)	
Assessment	The course assessment is an individual assignment: writing a policy note on CBA. The retake of the assignment is only for students who submitted their assignment in the first attempt and NOT for new or existing students who did not previously submit their assignment in the first attempt.	
Special Information	In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: take-home individual assignment.	
Category	MSc level	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Ethics Courses Must choose at least 1 out of these 4 courses

TPM003A	Water Ethics	5
Module Manager	Prof.dr.mr.ir. N. Doorn	
Co-responsible for assignments	Dr.ir. U. Pesch	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Water is essential for life. While we can in principle survive weeks without food, without water, we would not survive more than a couple of days. The aim of this course is to provide an introduction to philosophical water ethics. It will present the main philosophical issues in the water domain in a practical way by situating these issues within real cases. Questions addressed will include: Is it fair to drink water, knowing that the production of one cup of coffee requires 140 litres of water on average? Should we prioritize between different water uses, like agriculture, navigation, and recreation? Should we continue building hard flood defences at the expense of the environment? Should local citizens be given a voice in water-related decision making? These and other questions will be linked to ongoing discussions in the relevant other disciplines within philosophy, most notably in ethics of technology, philosophy and ethics of risks, environmental ethics, climate ethics, and global justice. This course will provide students with an understanding of approaches in water engineering and water policy. Students will get experience with discussing and evaluating ethical considerations in water engineering and water policy. Topics covered include: water and justice, water and economic valuation, water and human rights, water and responsibility, and water and engineering.	
Study Goals	At the end of this course you will be able to: Describe the world's water challenges, also in view of climatic and geographic developments. Describe and analyse the different water uses and explain how they may lead to competing demands; Describe how the world's water challenges pose questions of justice and apply different concepts of justice to concrete water problems; Recognize and critically reflect on the methodological and ethical aspects of economic valuation in the context of water; Describe and apply different legal frameworks and rights concepts to the context of water; Describe and apply different concepts of responsibility to the context of water; Recognize and analyse the value-ladenness of water engineering and design; Describe and analyse current water governance practices and identify opportunities to include ethical considerations in water governance; Write a well-argued essay on a topic related to water ethics.	
Education Method	The structure of classes will combine lectures with seminars and debates. Dependent on the Corona restrictions, the course may take place on campus or online.	
Books	"Water ethics: An introduction", by Neelke Doorn. Publisher: Rowman & Littlefield. 2019. ISBN: 9781786609519 (Pb) / 9781786609526 (E-book).	
Assessment	Group and individual assignments. The form of assessment is not dependent on whether the course will be delivered online or on campus.	
Enrolment / Application	The course TPM002A is the 3 ECTS version of the water ethics course. There is also a 5 ECTS version of the water ethics course, with course code TPM003A. Students enrolled in the 3 ECTS course follow exactly the same study activities as the students enrolled in the 3 ECTS course, except that they don't write the additional essay that students enrolled in the 5 ECTS version of the course need to write as one of their final assignments. This essay is worth 2 ECTS. All students should enrol in the 5 ECTS version of the course, also the students who do not wish to be eligible for this additional 2 ECTS. In the first week of the course, students can indicate whether they wish to follow the 3 ECTS or the 5 ECTS version of the course. There is no separate active Brightspace site for the 3 ECTS course (TPM002A). Please enrol for the course via the Brightspace site of TPM003A, also if you want to enrol for the 3 ECTS version of the course. Enrolment is possible until 1 November of the academic year. On that day, the total number of available places is determined. After 1 November, enrolment takes place upon approval and may be denied if there are no more places available.	
Special Information	Attendance of the tutorial sessions is mandatory. Students are allowed to miss only one session and only if they have a valid reason.	
Remarks	For ethics teaching at TU Delft, check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Sustainability Water Engineering Water management	
Targetgroup	All students enrolled in a MSc-program at TU Delft with an interest in water ethics.	
Category	MSc level	

WM0312CIE	Philosophy, Technology Assessment and Ethics for CIE	4
Module Manager	Dr. S.M. Copeland	
Instructor	Dr.ir. U. Pesch	
Instructor	E. Yaghmaei	
Instructor	S. Steinert	
Responsible for assignments	Dr. S.M. Copeland	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module explores philosophical and ethical questions in civil engineering, as well as providing a basic introduction into Technology Assessment.	
Study Goals	After having completed the course you: - know the basic philosophical issues in engineering knowledge; - can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. - are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	Virtual learning environment, introductory lecture and small number of question hours; mandatory assignment (more information will be provided during the introductory lecture and/or one of the question hours)	
Literature and Study Materials	- Ibo van de Poel and Lambèr Royakkers (2011). Ethics, Technology, and Engineering: An Introduction. Wiley Blackwell, West Sussex (ISBN 978-1-4443-3094-6 (hardcover) / 978-1-4443-3095-3 (paperback). - Additional reading material - Virtual learning environment (Brightspace).	
Assessment	Exam + mandatory assignment	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Tags	Ethics Philosophy	
Category	MSc niveau	

WM0329TU	Ethics and Engineering	6
Module Manager	Dr. F. Santoni De Sio	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	4/0/0/0 or 0/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	The first part of this course completely coincides with the WM320TU course. The WM329TU course is the WM320TU course plus the writing of an essay. Please check the WM320TU Brightspace for all relevant information about lectures, work sessions, exam, and the essay. You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: - can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. - are able to write an opinion article on ethical or societal aspects of technology, according to academic standards of writing - are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and interactive work sessions (including role playing sessions) concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	50%: grade for the participation in the WM320TU course, which consists of a written exam (60%), presentation, and active participation during the working group sessions (40%). 50%: grade of the essay	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the number of workgroups. Please enrol as soon as possible and in any case before the start of the course via Brightspace.	
Remarks	The course is run twice each year, starting in quarters 1 and 3. The first part of this course (concluded with the written test) coincides with the course WM0320TU.	
Elective	Yes	
Category	MSc niveau	

WM0376TU	Ethics of Technological Risks	5
Module Manager	Prof.dr. S. Roeser	
Responsible for assignments	Prof.dr. S. Roeser	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course investigates ethical aspects of technological risks. Debates about technological risks related to, for example, energy technologies, robotics and biotechnology frequently culminate in stalemates. This is due to the complexities and intricacies inherent to such debates as they involve scientific information and uncertainties, as well as ethical and emotional considerations. Conventional, quantitative approaches focus primarily on statistical information to risk. They do not explicitly incorporate important ethical considerations such as justice, fairness and autonomy. Emotions such as compassion, care and feelings of responsibility can draw attention to such ethical aspects of risky technologies. Taking emotions and ethical considerations seriously can lead to more fruitful deliberations between different stakeholders in which relevant concerns are taken seriously and are explicitly reflected upon. This course will study how approaches to ethical aspects of risk can lead to more morally responsible decision making and design of technological innovations.	
Study Goals	This course will provide students with an understanding of approaches to risk ethics and students will get experience with discussing and evaluating ethical aspects of risk.	
Education Method	Self-study, lectures and tutorials. Attending lectures and tutorials is obligatory.	
Assessment	On campus written exam (60% of the final grade) and essay (40% of the final grade)	
	In case of unforeseen circumstances, eg related to COVID-19 measures, the on campus exam may be replaced with an online (eg take home) exam	
	[There is also a 3 ECTS version of this course, i.e. WM0375TU; the only difference is that for the 3 ECTS version students do not write an essay.]	
Special Information	Contact person for additional questions: S.Roeser@tudelft.nl	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Philosophy	
Category	MSc level	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME General Electives

CME2100-11	Research Internship	10
Responsible Instructor	M.G.C. Bosch-Rekvelde	
Contact Hours / Week x/x/x/x	n.a	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This research internship focusses on practical work experience in day-to-day practice of civil engineering companies or institutes (contractors, consultancies, government, non-governmental organizations, etc.) in the Netherlands or abroad. As the title suggests: (a major part of) the practical work is dedicated to an individual (research) assignment, to be agreed upon with the internship coordinator and TUD subject supervisor, following the steps of the internship manual.	
Study Goals	The main objectives are * to develop your general engineering skills, * to develop your research skills, * to learn how to apply your technological know-how, * to put into practice any social and communicative skills you might have, * to gain a more complete insight into your own particular preferences and capabilities.	
Education Method	Practical work experience for a period of 8-10 weeks, focused on a specific research assignment. Before the student starts the internship, a brief project plan should be written and approved by the CME internship-coordinator (see course manual for the specific steps to be followed).	
Literature and Study Materials	On the Brightspace CME page you will find a digital version of the current course manual. This manual contains (amongst others) the information about the formalities that need to be arranged before the start of your internship. Any questions: please ask Mrs. Sandra Schuchmann (secretary, room 6.49).	
Assessment	The Internship is evaluated based on a written report to be handed in within 2 weeks after finishing the internship. An oral evaluation will follow with the students, the TUD subject supervisor and the coordinator. Also an evaluation of the company supervisor is required, see the course manual (Appendices) for all required forms. The report can be rejected if the student has not met the requirements as mentioned in the overview, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for rewriting the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out. -- Your grade for the internship is based on a judgement of the report on: 1) General engineering skills, 2) Technical knowledge, 3) Social and communication skills / reporting style, 4) Insight into own aptitude and affinity. The grading form can be found in the internship manual.	
Remarks	Finding a suitable internship is an important part of the course. Students should make sure to start organizing their internship well in time, preferably around 3-6 months before the planned date of departure! For information about the 'steps to be taken' please refer to the course documents (Course manual on CME Brightspace page).	
Expected prior Knowledge	Before starting the internship, you have finished at least all your obligatory courses of the CME program, since you need to reflect on the link between theory & practice. Preferably you also passed your electives. Typically, the internship is one of the final courses to complete (apart from the graduation project).	
Academic Skills	Important academic skills of thinking, writing and reflecting will be developed in the internship. The report is to be written in English.	
Literature & Study Materials	The study material consists of the (digital) course manual which is provided via Brightspace, Sandra Schuchmann or the internship coordinator (Marian Bosch-Rekvelde). All literature you have studied in your CME program can be relevant for your internship project: this obviously depends on the specific topic you will choose.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Permitted Materials during Exam	Not applicable	
Collegerama	No	

CME4900	Open Design Contractor Project	10
Responsible Instructor	Ir. R. Binnekamp	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This course is a multi-disciplinary project based on deepening or broadening some of the engineering & management system (EMS) concepts of their interest, see https://odesys.nl/ (the creative commons: MitC, 3C-Plan, Preferendus etc.). These EMS concepts are open design systems for decision support within Engineering Asset management or Construction Projects management or Service Operations management. Offshore and/or Civil Engineering students a EMS Crossover The goal of this crossover course for civil and/or offshore engineering students is to learn about engineering asset management systems which are a group of concepts/tools and methods to either support the primary in-service operations management processes or the construction projects management processes for new delivery of engineering assets. This allows CE/ODE students to broaden their horizon within an integrative systems engineering and management context that is connected to their specific engineering domain of interest. Rather than only focusing on what to engineer, this crossover focuses especially on how to manage the project and operations management processes. This crossover is preferably directly linked with an contractor context. CME students a preparatory R&D project The goal of this course for CME students is to carry out a mini grassroot R&D project by applying one of the known engineering asset management systems concepts out of other ODL courses. This allows CME students to deepen their existing EMS knowledge and thereby prepare well for the final MSc project. Students can choose between the following two options: Option 1: A Crossover or R&D project within an international contractor practice on developing a self-chosen EMS concepts for contractor projects and/or services management (CoI). This development project is then directly linked with the course concepts of CME 4120 Information Systems, CME 4481 Systems Engineering Design and CME4701 Contractor Management Systems. Option 2: A Crossover or R&D project within an international service provider (SPoI) on developing a self-chosen EMS concepts for engineering asset management quality of services provisioning. This development project is then directly linked with the course concepts of CME 4300 Engineering Asset Management.	
Study Goals	1. Development and reflective practicioning of engineering management systems concepts: Engineering Asset management or Construction Projects management or Service Operations management (i.e., strategic asset management plan/ project delivery planning/ in service operation plan) . 2. Familiarizing with integrative learning and intercultural aspects within an international contractor organization. 3. Further enhancement of practicioning with R&D concepts as a preparation for a full engineering construction or asset management oriented graduation thesis within a ODL setup. 4. Demonstrate and develop the personal outcome of the aforementioned learning goals, by internalizing the openings/ learning achievements in an original ODL project or crossover response.	
Education Method	The course is based on the Open Design Learning concept (ODLc), see https://www.open-design.school/ . For this course the ODL response is descibing the process and results of this Full-time project work.	
Assessment	The deliverable of this course is the so-called Open Design learning response. This ODL response is a (group) deliverable based on the self-chosen EMS concept of interest. The Open Design Learning (ODL) response is an original enabler demonstrating both the group and personal learning and development achievements. The ODL response illustrates how the EMS concept of interest has been linked and evaluated to the self-chosen CoI/ SPoI using a: 1) logical review and/or diagramming 2) computer model. All of these (incl. the open glass box model) should be presented in a self-chosen format such as report/ elaborative presentation/ digital audio or video files/ animation/ website. The ODL must contain a clear justification of the individual contributions of each group member. Each group member must also write an individual Comment in which he/she writes a collegial review of at least 1 concept of the ODL response for which he/she was not responsible. Good collegial Comments make use of specific ODL Commendation aspects (see next section). The Open Design Learning Commendation principle will be applied as a grading rubric for the ODL response. Both the CoI/SPoI content characteristics, and the students learning process are integrated within these commendation principles. We call it commendation because when we grade your response, we start from a grade of 10 and only deduct points if aspects are missing/only partially worked out. There are 5 commendation categories: Connect/ Construct /Conclude & Conspect /Convey /Convince	
Expected prior Knowledge	One of the following courses : CME 4300, CME4481, CME4701, CME5030 for E&S	
Academic Skills	Skills to: curiously connect to a committed system of interest; critically convince others with a compelling response; clearly convey a communicative response; creatively convert concepts into a comparative response; construct a properly compiled open glass box model; contextualize in a combined and open space; comprehensively conclude and cogently conspect your open ended solutions.	
Literature & Study Materials	Reference material of the following courses : CME 4300, CME4481, CME4701, CME5030 for E&S	
Judgement	After handing in the ODL response you will receive your grade using the ODL commendation principle. To pass the course your ODL grade should be higher than or equal to 6. After commending your ODL response one plenary open dialogue session (max. 3 hours, somewhere between week 12 and 14 to be announced) with one of the constructors will be scheduled. Only	

	during this session(s) we can reflect on your ODL response: bear in mind, the outcome of this reflection can result in a lower/equal/higher grade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

CME4950	CME Research and Development Project	10
Responsible Instructor	Ir. J.S. Hoving	
Instructor	M.G.C. Bosch-Rekvelde	
Instructor	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	x/x/x/x variable	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The student will, individually or in a team of multiple students, work on a CME research or development project which is part of ongoing CME research. This can be in collaboration with and under supervision by any CME staff member, including PhD's and postdocs. The research project can also be conducted in an affiliated research group (e.g. Deltares, TNO, research laboratory) or in a company R&D department. The CME research or development project provides students with the opportunity to work in a research environment, to master research techniques and further develop scientific thinking, argumentation and skills. It is an excellent opportunity for students to find out whether they aspire a scientific career and/or PhD-position. The student has to set up an assessment committee which consists of at least two supervisors, where at least one of them should be a full, associated or assistant professor in one of the faculties involved in the CME Master programme, i.e. Architecture & the Built Environment (ABE), Technology, Policy & Management (TPM) or Civil Engineering & Geosciences (CEG). Additionally, the student must inform their specialisation coordinator of the topic of their CME R&D project and the composition of the assessment committee (note here that this does not mean that the specialisation coordinator must be part of the assessment committee). The student is responsible for the planning, management and execution of the CME R&D project, in coordination with the assessment committee. The format of the deliverables of the CME R&D Project is not fixed and are agreed upon in consultation with the assessment committee, but the CME R&D project is always concluded by a final session with the assessment committee, including a presentation of the R&D project and a Q&A-session. The assessment committee is responsible for the assessment and determines the grade only after consultation of all committee members. As the CME R&D project is worth 10 credits, the student is expected to spend approximately 280 hours, which equals 7 weeks of 40 hours if executed full-time. However, the CME R&D project may also be executed part-time and is not necessarily restricted to the regular education periods, pending the approval of the assessment committee.	
Study Goals	By completing the CME Research & Development Project, you are able to: - Familiarize yourself with fundamental research and developments related to the topic; - Delineate a research, development or design problem and formulate a methodology to address the research or development statement in a well-structured way; - Plan, execute and manage the project effectively within the available time and resources; - Demonstrate your understanding of the scientific, societal and ethical context of the CME R&D Project; - Verify, interpret, and critically reflect on obtained results; - Draw relevant conclusions and give useful recommendations; - Present the CME R&D Project in a well-structured manner;	
Education Method	Self-study; support is available from the supervisors, i.e. the members of the thesis committee.	
Assessment	The assessment is based on: - the deliverables of the CME R&D project agreed upon with the assessment committee, - presentation of the CME R&D project, and - the Q&A-session after the presentation.	
Expected prior Knowledge	Any relevant courses in your individual CME study program. Especially the courses on Ethics and Methodology are of specific relevance.	
Academic Skills	Academic thinking and argumentation demonstrated by unequivocally communicating conclusions and recommendations, by addressing the knowledge, theory and motives, as well as societal and ethical considerations that underlie these.	
Literature & Study Materials	Relevant literature and study materials regarding the R&D topic are to be collected by the student from the open literature and may consist of research reports, peer-reviewed articles, books, proceedings and presentations.	
Judgement	Judgement by the assessment committee.	
Permitted Materials during Exam	Not applicable	
Collegerama	No	

TUD4040-10EC	Joint Interdisciplinary Project	10
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Module Manager	Ir. B.J.E. de Bruin	
Module Manager	Dr. R.G. Klaassen	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily) technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, , Embraer, Huisman, VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, KLM	
Study Goals	Meta-cognitive abilities attributable to interdisciplinary learning LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (ILO;K) Scientific and intellectual development LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I) Research and design capabilities LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J) Collaboration and communication in an interdisciplinary team LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K) Self-adjustment and reflection capabilities LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Course Relations	The course TUD4040-10 EC is the CME-version of the course TUD4040 Joint Interdisciplinary Project. While the workload of the JIP is principally worth 15 ECTS, based on the contents of the CME-curriculum, all CME-students already possess competences regarding interdisciplinary learning, collaboration and communication in interdisciplinary teams when they start this course. For CME-students therefore, the JIP only warrants and counts for 10 ECTS towards their 120 ECTS CME study program.	
Assessment	Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions: Problem statement review the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. Midterm review The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level. In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback) Summative assessment on module level Final review the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es). Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added. The minimum grade for the summative assessment is => 5.8. Process grade is 20% The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.	
Expected prior Knowledge	To enter JIP, a student should be a 2nd-year master student; have completed the first year of the masters degree programme and will enrol in the second year of the masters degree programme at TU Delft. be full time available (40 hours per week) in Q5; meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV	

Academic Skills	Students will be selected by the JIP core staff based on motivation, study progress and recommendation. Students work in a team on a complex problem provided and guided by a company (often on location) academic and JIP staff. Students are required to apply their disciplinary content knowledge, research/design/modelling skills, innovation, interdisciplinary teamwork, professional and personal development skills (including ethics and responsible engineering).
Literature & Study Materials	Literature is provided by academic staff and company based on the topics addressed.
Permitted Materials during Exam	n/a
Collegerama	No

WM0201TU-ENG	Technical Writing	2
Module Manager	Drs. J.A. Weitenberg	
Instructor	Drs. J.A. Weitenberg	
Co-responsible for assignments	Drs. A.E. Kam	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is geared to two groups of writers: engineers who need to write technical reports and scientists who need to write scientific articles. Modern engineers are required to possess excellent communication skills. Not only do they need to know their business, but they also need to know how to communicate this business to other experts, decision makers, and implementers. They need to be able to write accessible technical reports and proposals and to hit the correct tone in professional correspondence. The goal of scientific research is publication. Scientists are rarely measured by their knowledge of science; instead, they are measured and become known (or remain unknown) by their publications. Scientific articles need to be well structured to allow for efficient reading and accessible to the scientific community. They should also be written in a professional style, convincing, effective, and without language errors. Both groups will benefit from this course, which focuses on both the quality of the text and on the writing process itself. The following subjects will be dealt with: - a systematic approach to writing - requirements to technical reports and scientific papers and their components - structuring texts on several levels - writing convincingly - writing "proper and appropriate" language - referencing Seven lectures will be used for instruction, feedback, and practice. Students will be required to hand in several assignments while they are working on the technical or scientific report that needs to be written to complete the course. No more than one lecture may be missed.	
Study Goals	Improving written communication skills.	
Education Method	Workshop. Writing assignments, peer feedback.	
Literature and Study Materials	Elling et al (2011): Report Writing for Readers with Little Time. 1st edition, Noordhof Uitgevers. All other materials are available in Brightspace.	
Assessment	Several smaller assignments and one final report or scientific paper.	
Enrolment / Application	Brightspace	
Elective	Yes	
Category	MSc level	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Thesis

CME5100	CME Master Thesis Preparation	5
Responsible Instructor	Ir. J.S. Hoving	
Course Coordinator	M.G.C. Bosch-Rekvelde	
Course Coordinator	Ir. R. Binnekamp	
Course Coordinator	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	All CME students of cohort 2020-2021 and after, in preparation of their Master thesis project.	
Summary	CME5100 Master Thesis preparation is concluded by approval of the thesis proposal and always succeeded by CME5200 CME Master Thesis (30 credits). The Master Thesis preparation is not bound to a certain period.	
Course Contents	The students will prepare their Master thesis by executing the following steps: 1. Formulate a rough research, development or design proposal in the field of your Construction Management and Engineering specialisation* (max. 1 A4). This proposal may be based on a) the ideas of an involved company, b) university research, or c) personal ideas. 2. Submit your rough proposal to one of the graduation coordinators (preferably the coordinator that best fits the topic) and get approval to continue the thesis preparation. 3. Find a main thesis supervisor in consultation with the graduation coordinator. 4. Establish a thesis committee in consultation with the main thesis supervisor and, if desired and not already involved, find a company/organization to execute the Master thesis with. 5. If you choose to graduate at a company, arrange a graduation agreement in which all conditions for the graduation are stipulated and agreed upon by the company, the student and TU Delft. Note that the rough proposal from step 1 must be included in the graduation agreement. The standard TU Delft graduation agreement, the corresponding procedure and its conditions are available on the Faculty CEG student portal. 6. Finalize the full research, development or design proposal, including a) introduction of the thesis topic, b) problem description and definition, c) thesis objective, d) methodology and e) planning of the Master thesis project. The course is concluded by approval of the thesis proposal during the Master thesis kick-off meeting. Note here that without a graduation agreement signed by all parties involved, approval of the kick-off proposal may not be given. *) The thesis topic must be chosen within the scope of your CME specialisation, which is either Design & Integration (D&I), Engineering & Systems (E&S) or Projects & People (P&P). Each specialisation may have specific requirements regarding the thesis topic: please check the CME Brightspace community for the corresponding information.	
Study Goals	Formulate a thesis topic, establish a thesis committee, consisting of at least a main supervisor and a chairman, arrange a graduation agreement and write a thesis proposal to be approved by the thesis committee at the Master thesis kick-off meeting.	
Education Method	Self-study; support is available from the graduation coordinators and/or prospective thesis supervisors.	
Prerequisites	Students may start the graduation, and thereby the course CME5100 Master Thesis Preparation, if they have successfully completed all courses of the first year of the Master CME, as well as all courses of the first quarter of the second year of the Master CME nominally, and thus within the minimum amount of study time possible. Students who have not successfully completed all courses of the first year and the first quarter of the second year of the Master CME nominally, may start the graduation, and thereby the course CME5100 Master Thesis Preparation only if all of the following requirements are met: a. The student has completed the synchronisation course, b. The student has completed the course CIE4030 Methodology for Scientific Research, c. The student has completed at least one of the Ethics courses available in the CME curriculum, d. The student has completed at least 76 credits worth of courses out of the minimum 84 credits required for the study programme, excluding the graduation. This means that the student has at most 8 credits worth of courses to complete during the graduation. Admission to start the graduation and this course may be granted while still awaiting exam results to fulfil above requirements, but only by request and based on a clarification by the student accompanying the submission of the CME-1 form (see Enrolment/Application).	
Assessment	Approval is given by the thesis committee after assessment of the Master thesis proposal and the presentation of the thesis proposal at the Master thesis kick-off meeting. Formal approval of the thesis proposal is given by submission of the form: 'CME-2: Graduation Kick-off Proposal and Committee Approval' available at the Faculty CEG student portal.	
Enrolment / Application	You must ask permission to start the graduation, and thereby start CME5100 Master Thesis Preparation, using the form: CME-1: Admission to start Graduation Thesis available at the Faculty CEG student portal.	
Contact	Coordinators: Dr. Ir. Marian Bosch-Rekvelde Coordinator Projects & People m.g.c.bosch-rekvelde@tudelft.nl Dr. Daan Schraven Coordinator Design & Integration d.f.j.schraven@tudelft.nl Dr. Ir. Ruud Binnekamp Coordinator Engineering & Systems r.binnekamp@tudelft.nl CME Secretary: Mrs. Sandra Schuchmann Building 23 (CiTG), room 6.49, tel: +31 (0)15 27 84774 s.c.m.schuchmann@tudelft.nl	
Expected prior Knowledge	All courses in your individual CME study program. Especially the courses on Ethics and Methodology of Scientific Research are of specific relevance for the process during the	

Academic Skills	CME Master Thesis.
Literature & Study Materials	Academic thinking and Academic writing skills. See course CIE4030 Methodology for Scientific Research. Additionally, you can benefit from: - Creswell; Research Design: Qualitative, Quantitative, and Mixed Methods Approaches, - Maylor, Blackmon & Huemann; Researching Business and Management, - Roozenburg & Eekels; Product Design, Fundamentals and Methods.
Judgement	Judgement by the thesis committee.
Permitted Materials during Exam	Not applicable
Collegerama	No

CME5200	CME Master Thesis	30
Responsible Instructor	Ir. J.S. Hoving	
Course Coordinator	M.G.C. Bosch-Rekvelde	
Course Coordinator	Ir. R. Binnekamp	
Course Coordinator	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	All CME students of cohort 2020-2021 and after, as part of their graduation.	
Summary	CME5200 CME Master Thesis is always preceded by CME5100 Master Thesis preparation, which is concluded by approval of the thesis proposal by means of the CME-2 form available at the Faculty CEG student portal. The CME Master Thesis is not bound to a certain period. Further information regarding the CME graduation process is available at the graduation/specialisation coordinators. Possible information sessions regarding the CME Master Thesis are announced through the CME community Brightspace.	
Course Contents	The CME Master Thesis consists of individually conducting a research, development or design project in the field of your Construction Management and Engineering specialisation*. The thesis project is based on the thesis proposal approved by your thesis committee as part of CME5100 Master Thesis preparation. During the CME Master Thesis, you are to demonstrate the capacity to conduct your research, development or design project at the academic Master level by showing the ability to be able to independently plan, execute and manage your thesis project. During the thesis project you are additionally expected to demonstrate the ability to apply relevant theory, to set-up and execute a sound methodology, to critically interpret and reflect on obtained results and to present your findings in a well-structured manner through written and oral communications in proper English. As the CME Master Thesis is worth 30 credits, you are expected to finalise the thesis project in approximately 6 months. This excludes the time spent on CME5100 Master Thesis preparation. *) The thesis topic must be chosen within the scope of your CME specialisation, which is either Design & Integration (D&I), Engineering & Systems (E&S) or Projects & People (P&P). Each specialisation may have specific requirements regarding the thesis topic: please check the CME community Brightspace for the corresponding information.	
Study Goals	By completing the CME Master Thesis, you are able to: - Familiarize yourself with fundamental research and developments related to the thesis topic; - Demonstrate the capability to independently determine and apply relevant theory; - Delineate a research, development or design problem and formulate a theoretical and/or experimental methodology to answer the corresponding research question in a well-structured way; - Plan, execute and manage the thesis process effectively considering the available time and resources; - Produce results that are scientifically and technologically relevant; - Verify, interpret, and reflect on obtained results in a critical manner; - Critically reflect on the thesis project at the level of your peers; - Present the thesis project in a well-structured report, or in a draft scientific paper, written in correct English that includes the problem statement, methodology, scientific context, verification of applied methods, obtained results and critical reflection; - Present the thesis project in a well-structured oral presentation to a wide audience that next to your peers, may include audience not directly familiar with the thesis topic; - Defend your work, results and reflections in presence of your peers by placing the thesis project into its scientific context and by explaining the contribution to the field;	
Education Method	Self-study; support is available from the main supervisor and the other members of the thesis committee.	
Practical Guide	Throughout the thesis project, your main supervisor is your first point of contact and main source of supervision. During the CME Master Thesis, there will be several meetings with your thesis committee: multiple progress meetings, a green-light meeting and the final graduation session, consisting of a public presentation and the defence. You are expected to organize these meetings in consultation with, and only after the agreement of, your main supervisor. After each meeting has been scheduled, it is your responsibility to make sure that all members of the graduation committee are informed about the date, time and location of the upcoming meeting. Note here that any reservations of lecture or meeting rooms required are to be made by Mrs. Sandra Schuchmann (see contact info). Progress Meetings: During the graduation, there are generally 1 or 2 so-called progress meetings with the full graduation committee present. During these meetings, the student is expected to present his/her progress and to discuss the way forward. Not later than 1 week prior to each progress meeting, the student is requested to send a digital version of the draft report to all members of the graduation committee. Green-light Meeting: The purpose of the green-light meeting is to determine whether the student is ready for the final graduation session. Not later than 1 week prior to this meeting, a(n) almost final draft of the thesis is to be submitted to all committee members. During the green-light meeting, the student is expected to give a 20- to 30-minute presentation (excl. questions) giving an overview of his/her graduation project with a focus on results and conclusions. Based on this draft of the thesis and the presentation, the graduation committee will decide whether the quality of the work is sufficient to get the green-light. Upon receiving the green-light, the student is to submit the form CME-3: Green Light for Graduation Thesis Defence, available at the Faculty CEG student portal. Note here that the official rules (R&G Board of Examiners, article 9b) state that the CME-3 form must be submitted to the student administration no later than 20 working days before the graduation date. Therefore, the green-light meeting must always take place at least 4 weeks before the final graduation session. Here note that green-light for the final graduation session is formally only given after approval by the administration office that checks whether you meet all the program requirements, which means that you have successfully completed all courses that are part of your individual CME study program. Final Graduation Session: Not later than 1 week prior to the final graduation session, you are to submit both a hard-copy and a digital version of the final Master thesis to all members of the graduation committee. The final thesis is based on the final draft delivered prior to the Green-light meeting but includes additional content and improvements according to the agreements made during the green light	

	meeting. During the final graduation session, you are to give a public presentation focusing on your objectives, obtained results, conclusions and own contribution. In addition, the presentation should also be inspiring and understandable for the general public. After the presentation, you are to publicly defend your work by answering the questions and challenges posed by your thesis committee and critically reflecting on your own work. If the combination of the thesis, presentation and defence are deemed worthy by the thesis committee, the session commences with the diploma ceremony. If, during the course of your thesis project, the composition of your thesis committee changes, please submit the form CME-2a: Changes to Composition of the Graduation Committee, available at the Faculty CEG student portal.
Prerequisites	Students may start the CME Master Thesis only after completing the course CME5100 Master Thesis Preparation. To conclude CME5100 Master Thesis Preparation, the student has organised a Master thesis kick-off meeting during which the thesis committee has approved the thesis proposal, after which the student has submitted the form: 'CME-2: Graduation Kick-off Proposal and Committee Approval'. This is the starting point of your CME Master Thesis.
Assessment	The requirements for the course CME5100 Master Thesis Preparation, and thereby the requirements to start the graduation, are given in the course description for CME5100. The assessment of the Master Thesis is done on the basis of the final attainments of the CME program and takes place after the final graduation session, including the public presentation and public defence. This is done in a short, closed meeting of the thesis committee, without presence of the student. The assessment is performed by the university members of the thesis committee; external (company) members only have an advisory role. In the assessment, the following three main aspects are distinguished: 1. Content/Product: Quality of research, development or design presented in the Master Thesis; This aspect has to be sufficient in itself to lead to a sufficient overall score. 2. Process: Working and learning attitude during the thesis project. 3. Presentation and Communication: a) Report, b) Oral presentation and defence
Enrolment / Application	The student must ask permission to start the CME Master Thesis by submission of the form: CME-2: Graduation Kick-off Proposal and Committee Approval, and thereby successfully completing CME5100 Master Thesis Preparation.
Special Information	The following graduation forms exist and are available at the TU Delft student portal and at the CME-community on Brightspace: - CME-1: Admission to start Graduation Thesis - CME-2: Graduation Kick-off Proposal and Committee Approval - CME-2a: Changes to Composition of the Graduation Committee - CME-3: Green Light for Graduation Thesis Defence For more detailed information about the CME Master Thesis, its process and any possible additional requirements per specialisation please check the CME-community on Brightspace.
Contact	Coordinators: Dr. Ir. Marian Bosch-Rekvelde Coordinator Projects & People m.g.c.bosch-rekvelde@tudelft.nl Dr. Daan Schraven Coordinator Design & Integration d.f.j.schraven@tudelft.nl Dr. Ir. Ruud Binnekamp Coordinator Engineering & Systems r.binnekamp@tudelft.nl CME Secretary: Mrs. Sandra Schuchmann Building 23 (CiTG), room 6.49, tel: +31 (0)15 27 84774 s.c.m.schuchmann@tudelft.nl
Expected prior Knowledge	All courses in your individual CME study program. Especially the courses on Ethics and Methodology of Scientific Research are of specific relevance for the process during the CME Master Thesis.
Academic Skills	Academic thinking and Academic writing skills.
Literature & Study Materials	Relevant literature and study materials regarding the thesis topic are to be collected by the student from the open literature and may consist of research reports, peer-reviewed articles, books and proceedings.
Judgement	Judgement by the thesis committee.
Permitted Materials during Exam	Not applicable.
Collegerama	No

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Thesis for students who started before September 2020

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